

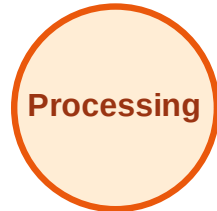
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



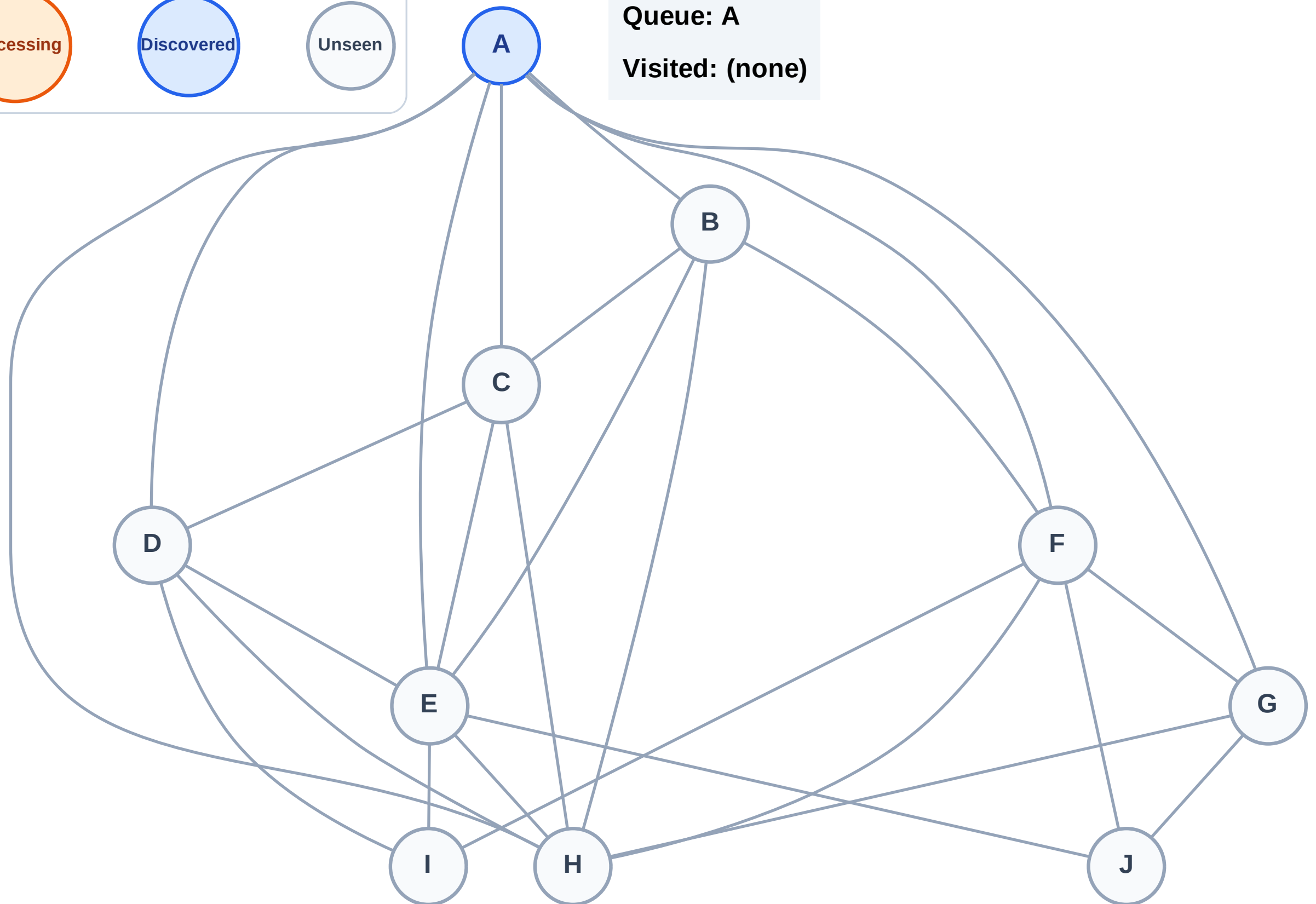
Discovered



Unseen

Queue: A

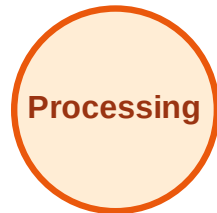
Visited: (none)



BFS Traversal

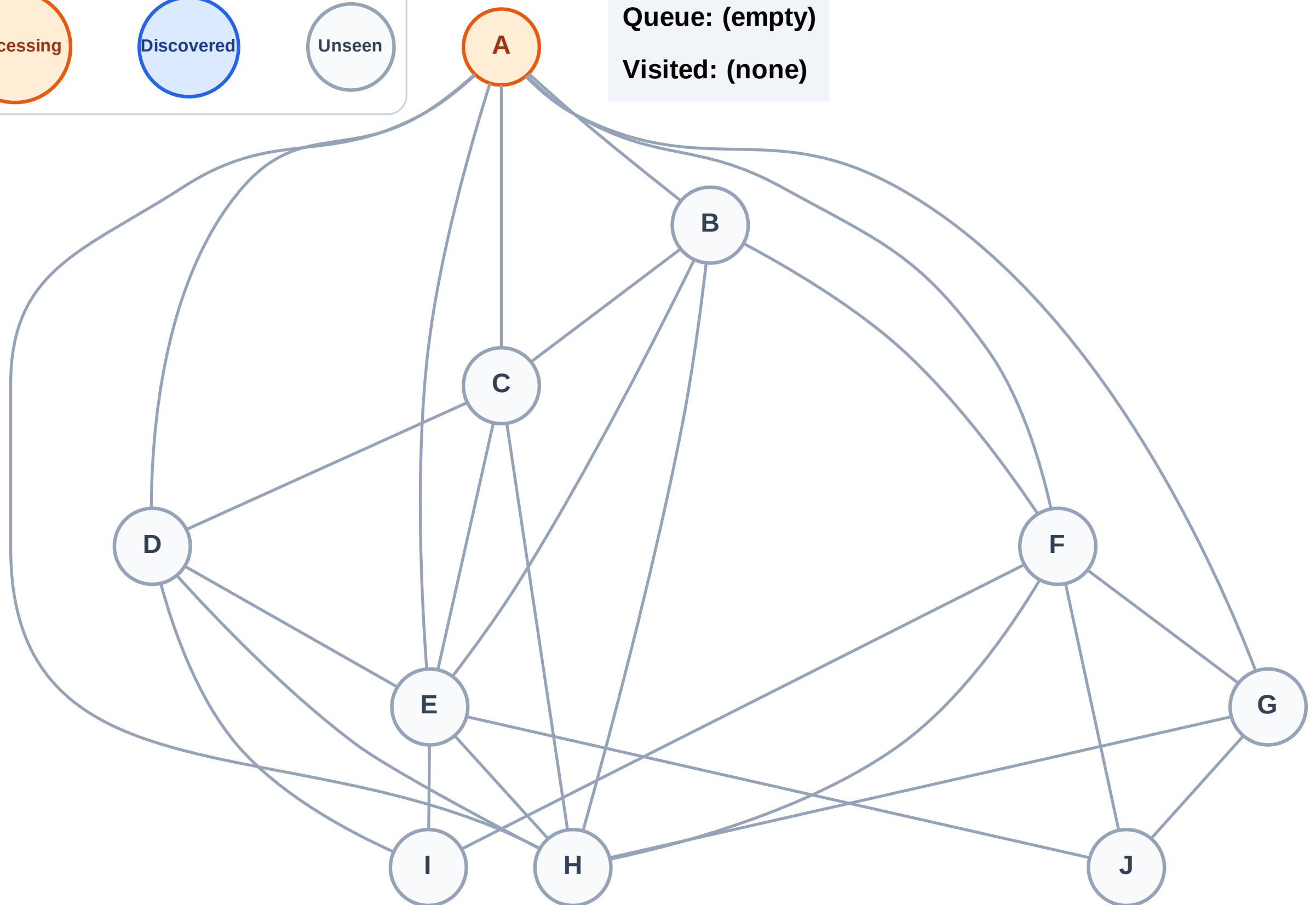
Step 2: Process node A

Legend



Queue: (empty)

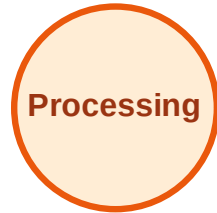
Visited: (none)



BFS Traversal

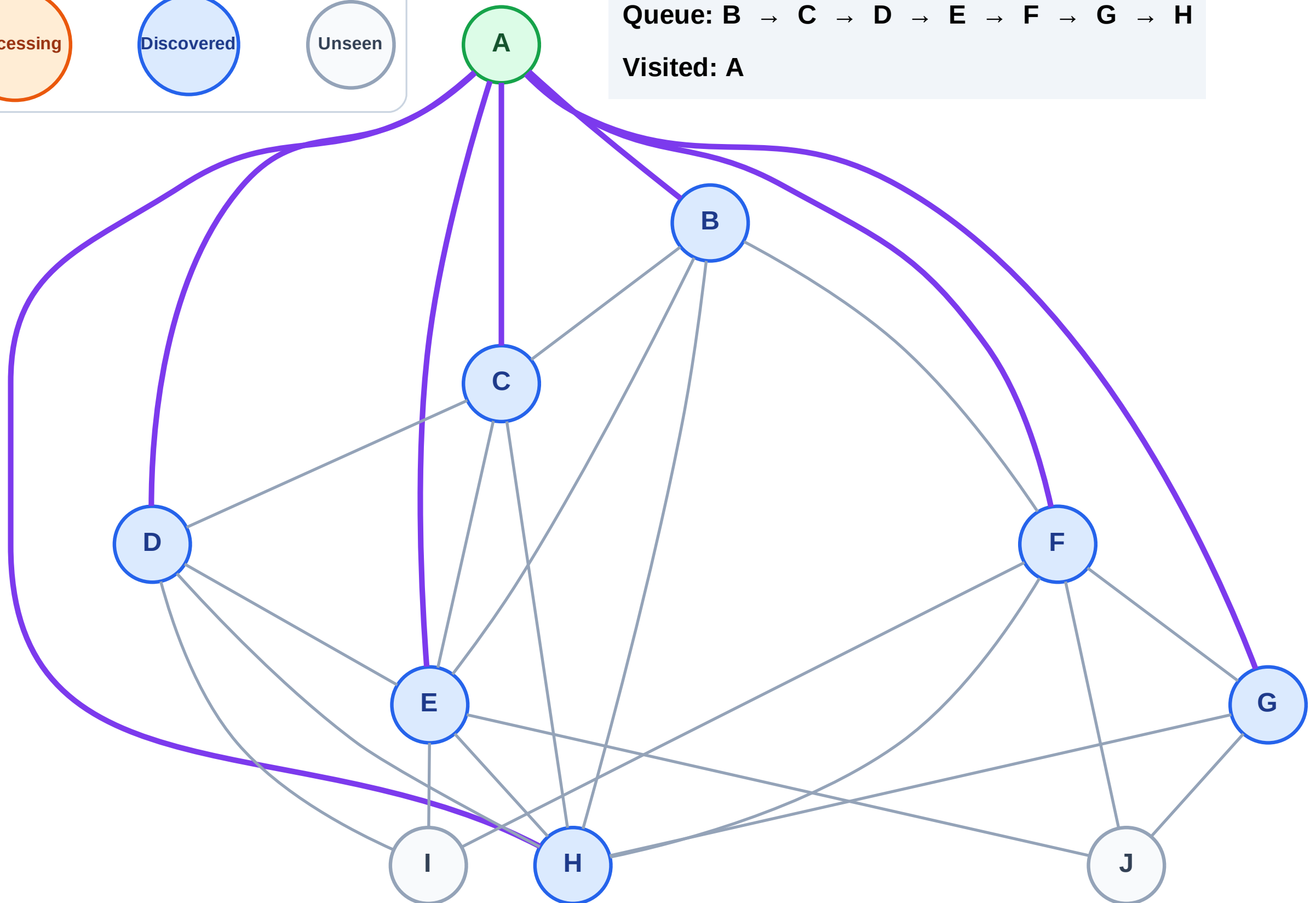
Step 3: Discover B, C, D, E, F, G, H from A

Legend



Queue: B → C → D → E → F → G → H

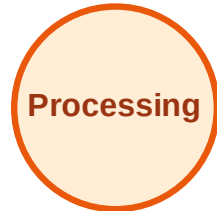
Visited: A



BFS Traversal

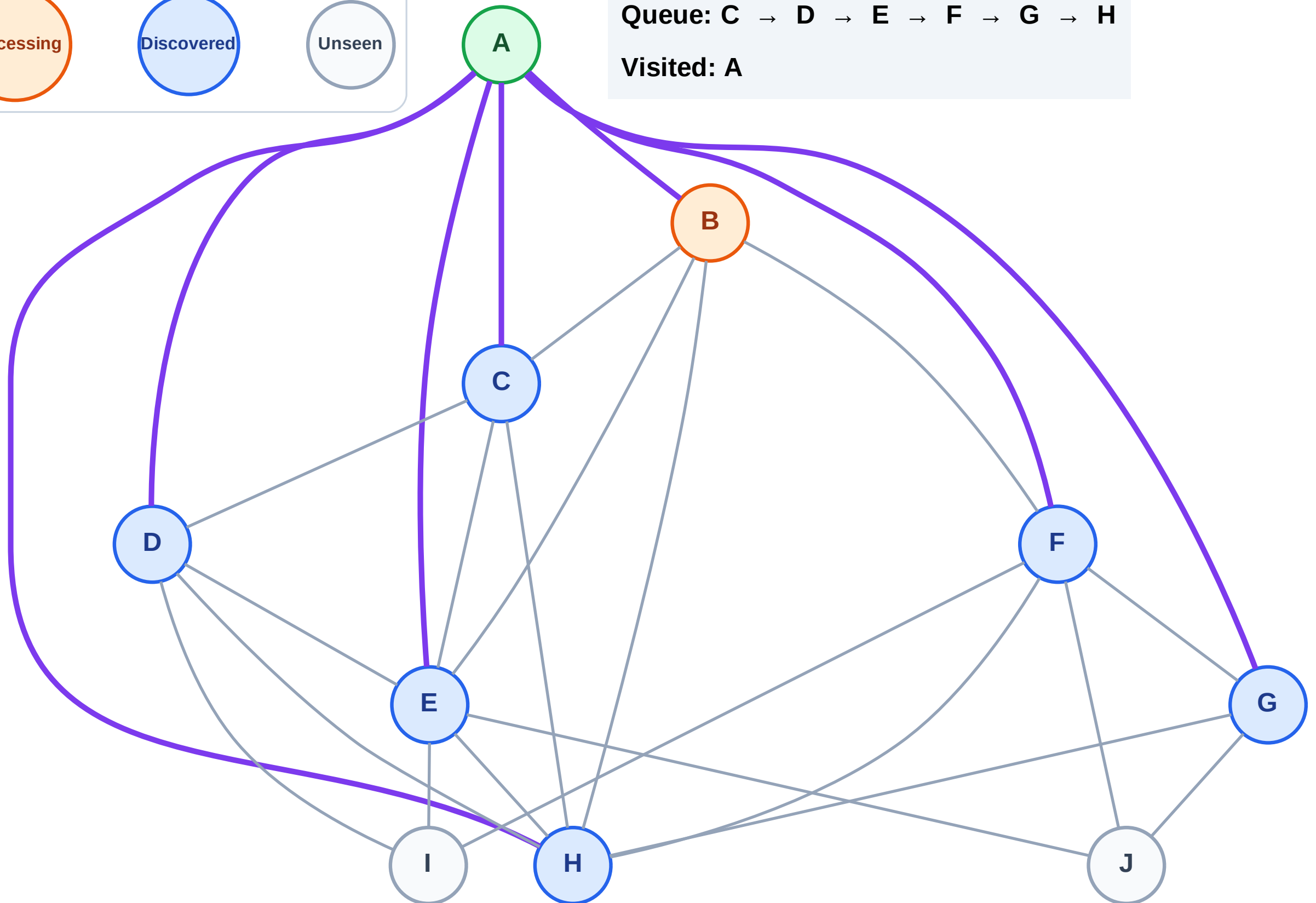
Step 4: Process node B

Legend



Queue: C → D → E → F → G → H

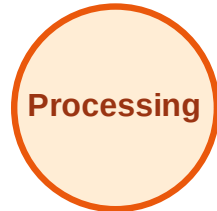
Visited: A



BFS Traversal

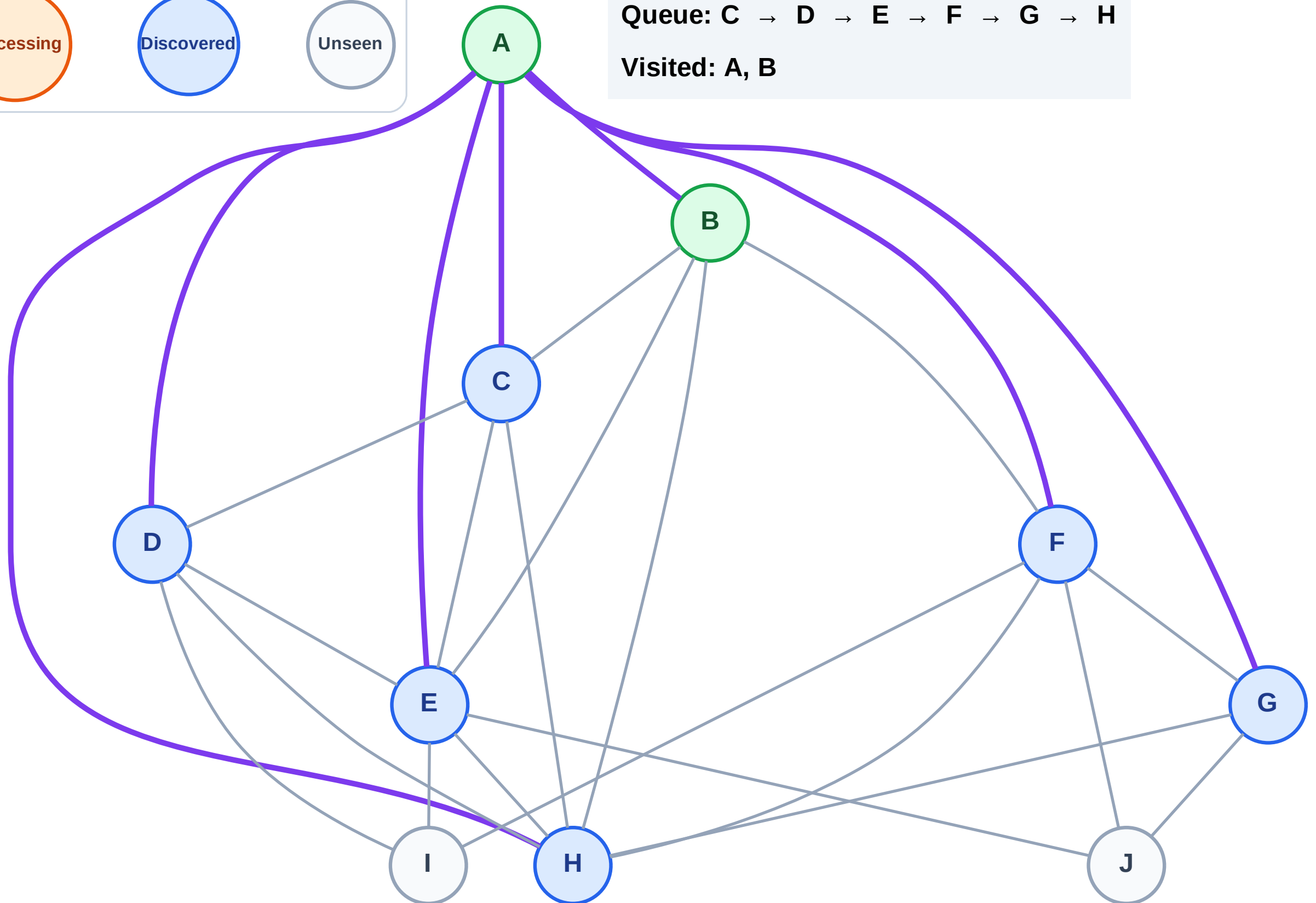
Step 5: No new neighbors from B

Legend



Queue: C → D → E → F → G → H

Visited: A, B



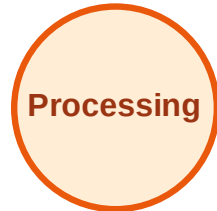
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



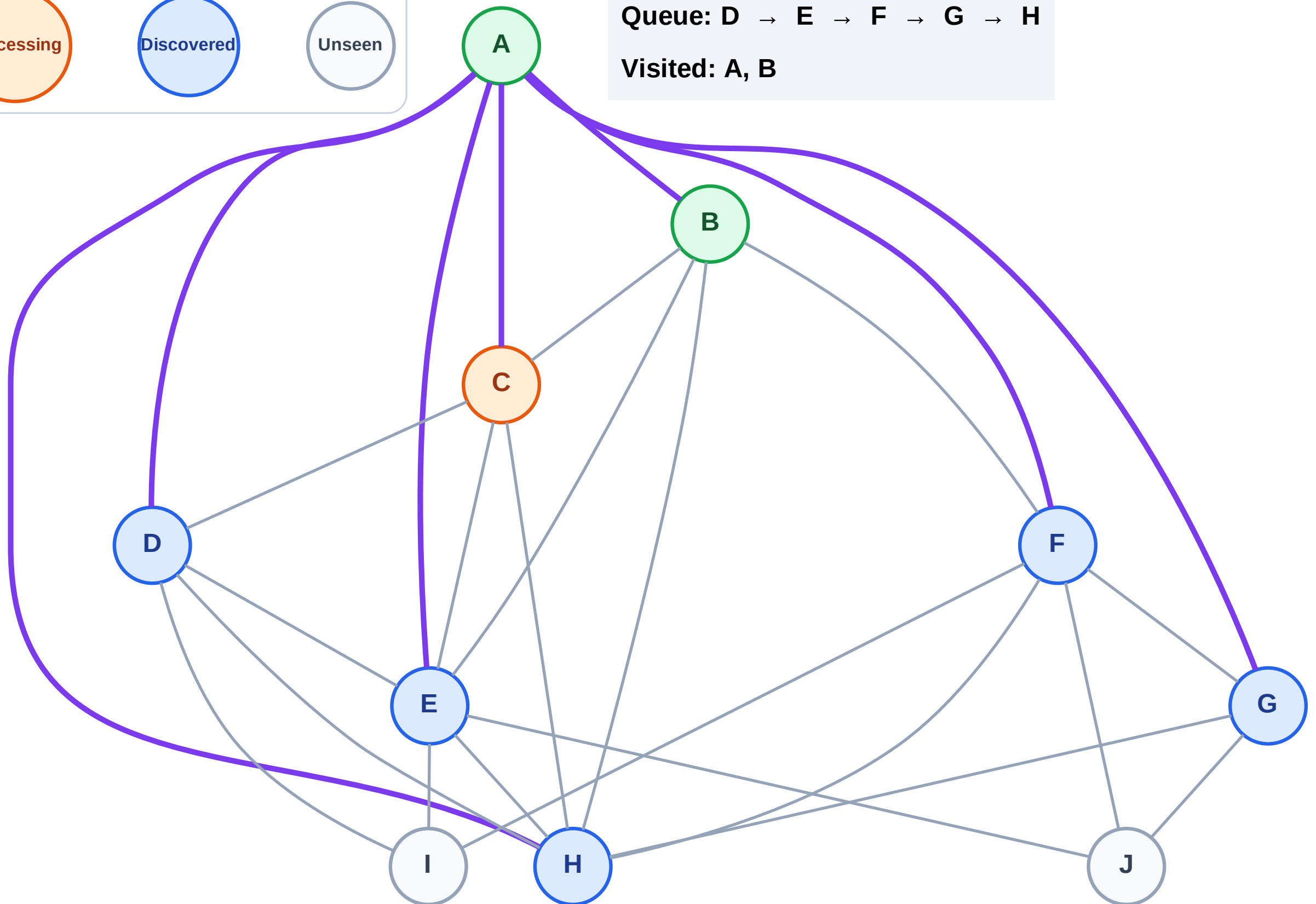
Discovered



Unseen

Queue: D → E → F → G → H

Visited: A, B



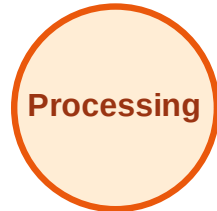
BFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



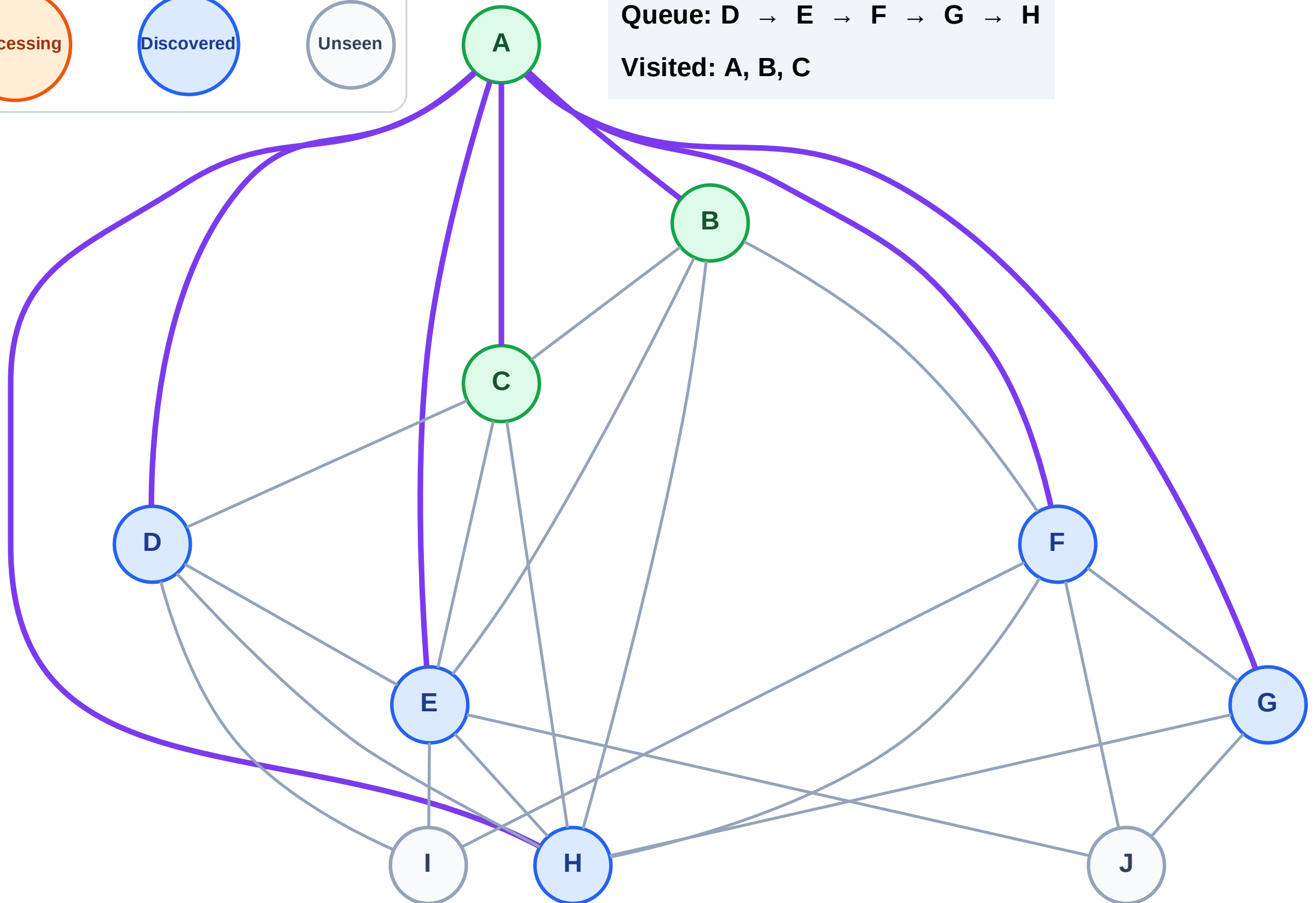
Discovered



Unseen

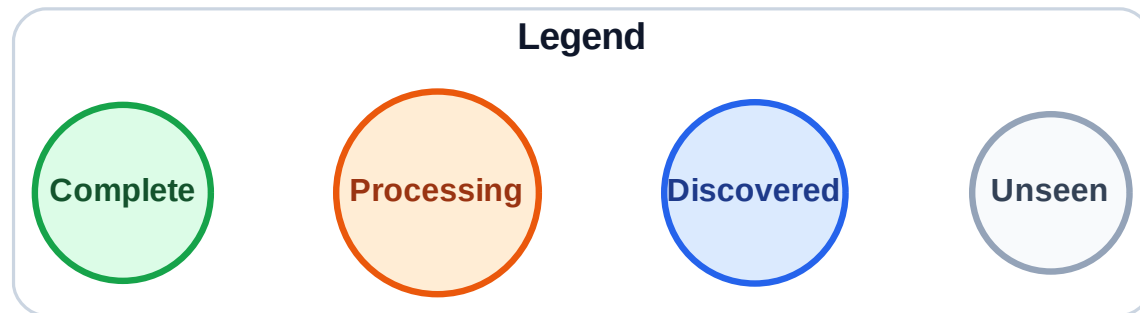
Queue: D → E → F → G → H

Visited: A, B, C



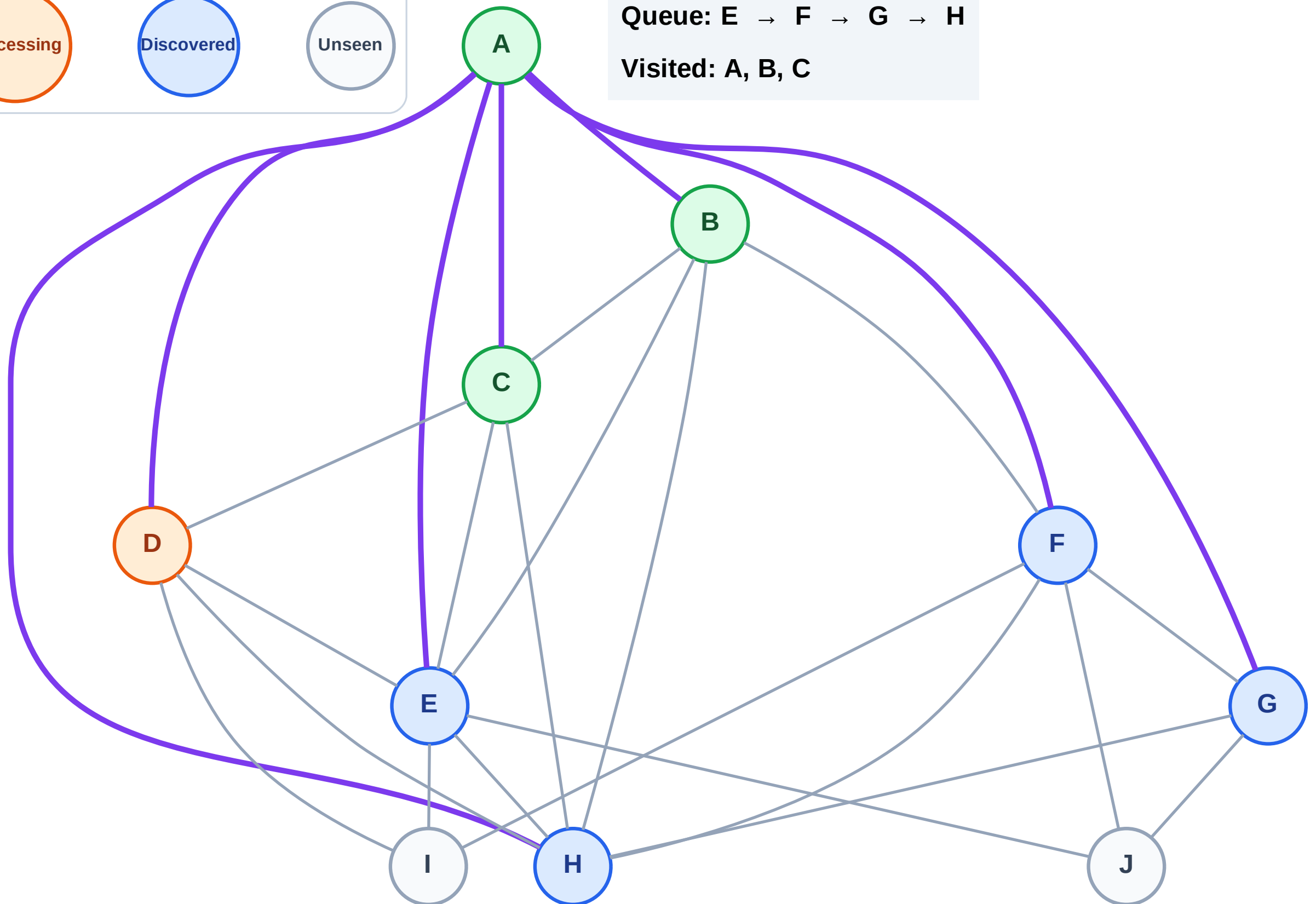
BFS Traversal

Step 8: Process node D



Queue: E → F → G → H

Visited: A, B, C



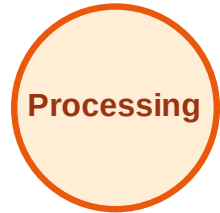
BFS Traversal

Step 9: Discover I from D

Legend



Complete



Processing



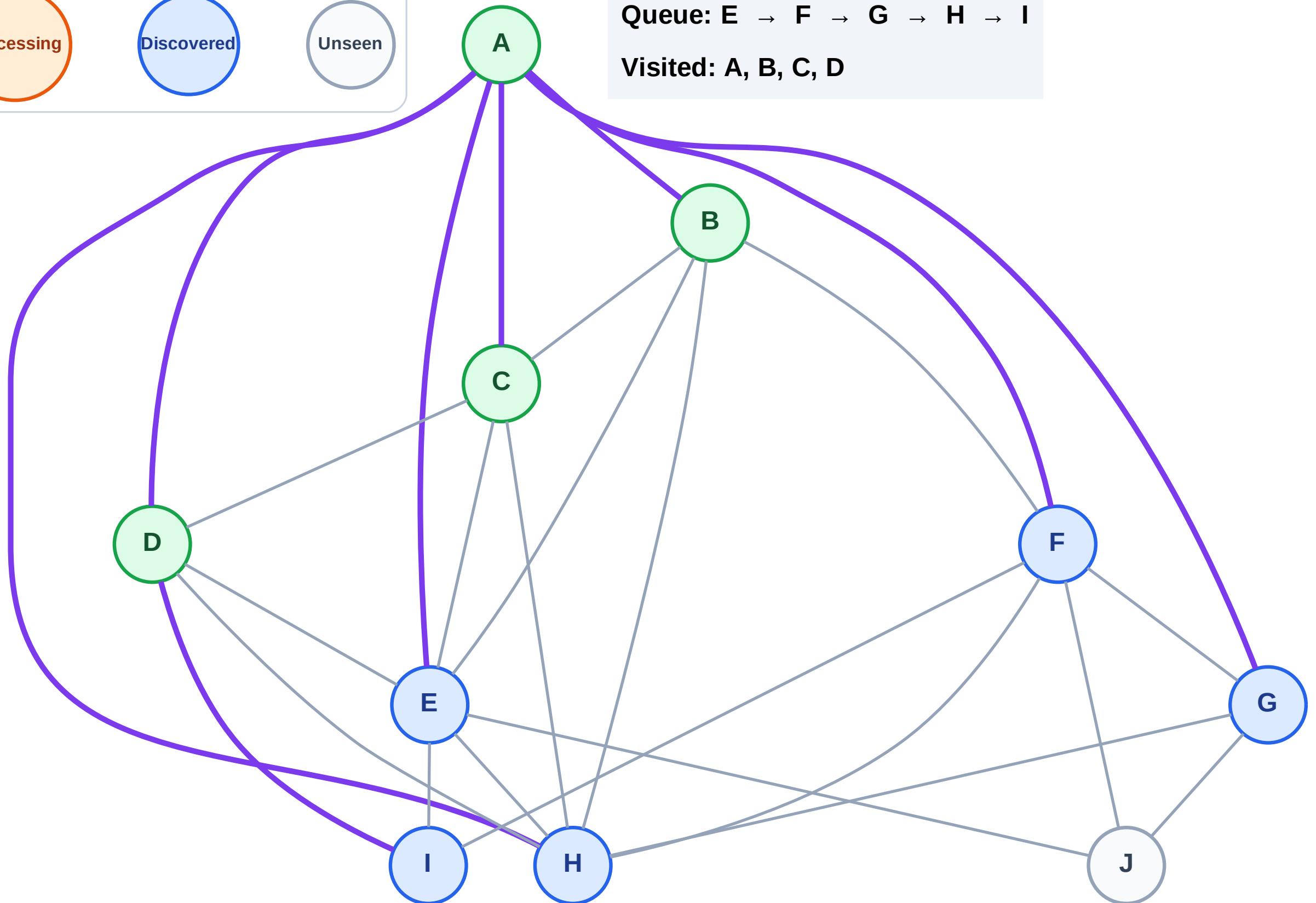
Discovered



Unseen

Queue: E → F → G → H → I

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

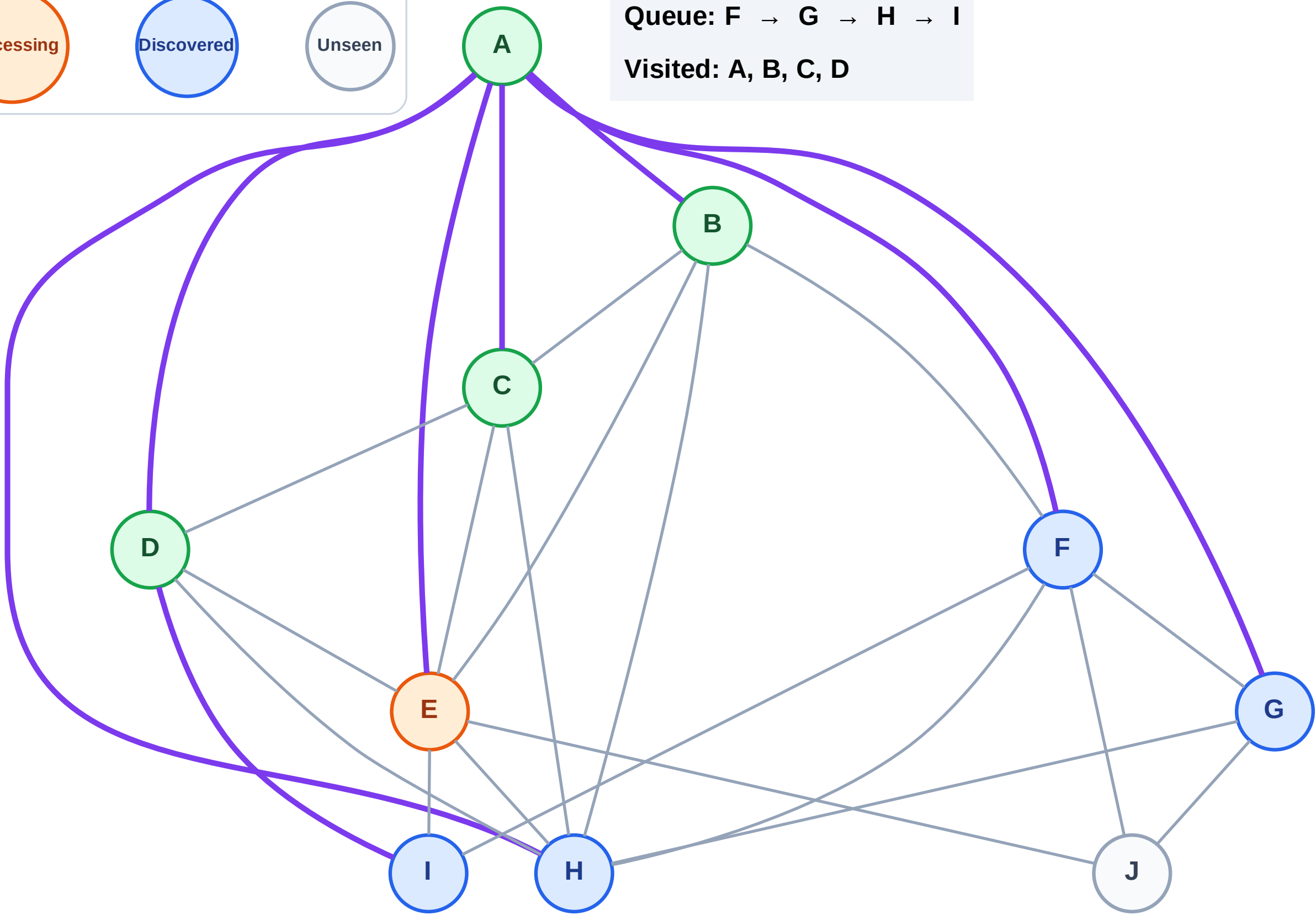
Processing

Discovered

Unseen

Queue: F → G → H → I

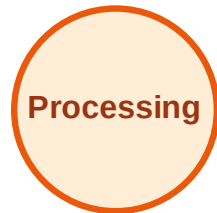
Visited: A, B, C, D



BFS Traversal

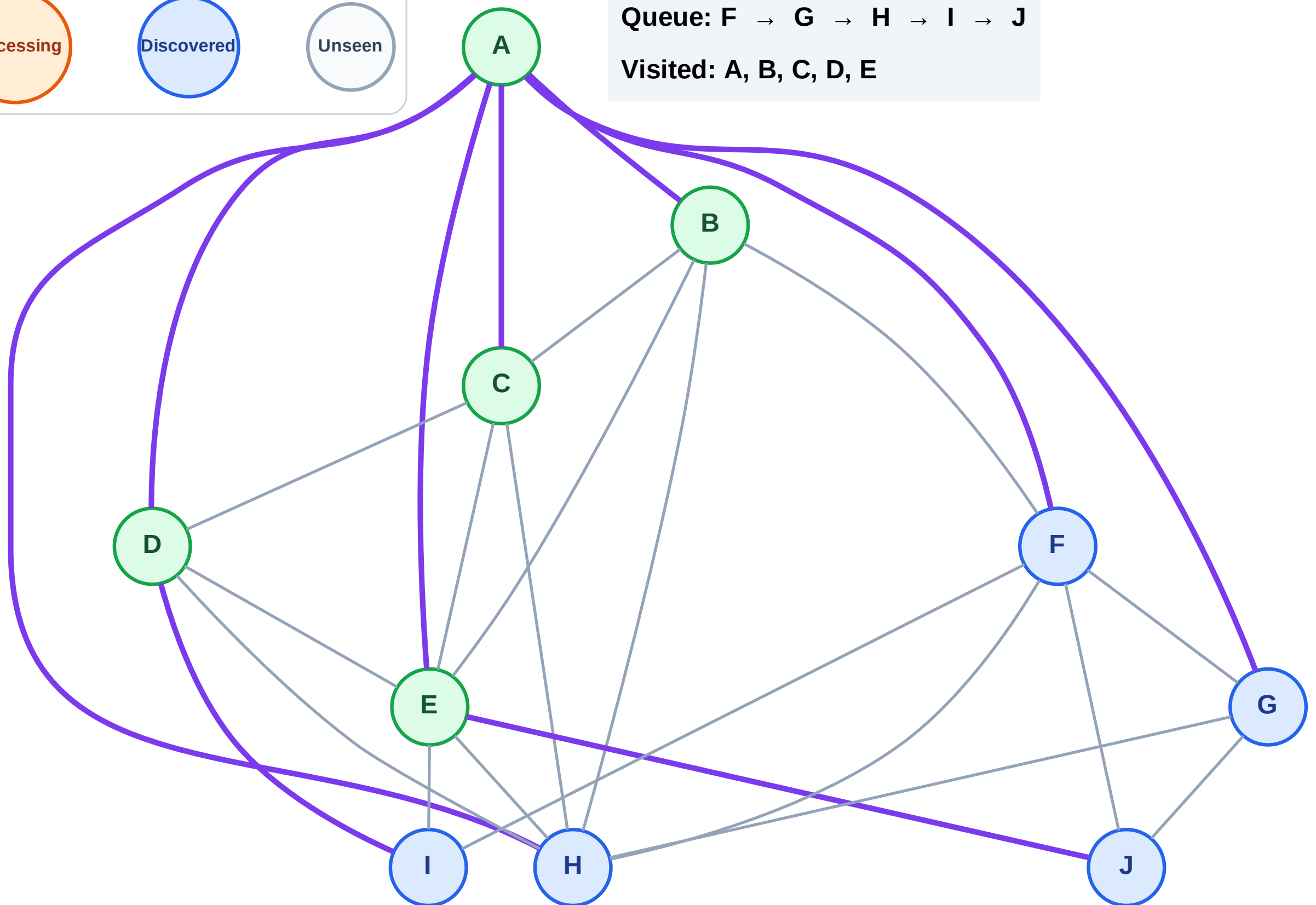
Step 11: Discover J from E

Legend



Queue: F → G → H → I → J

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node F

Legend

Complete

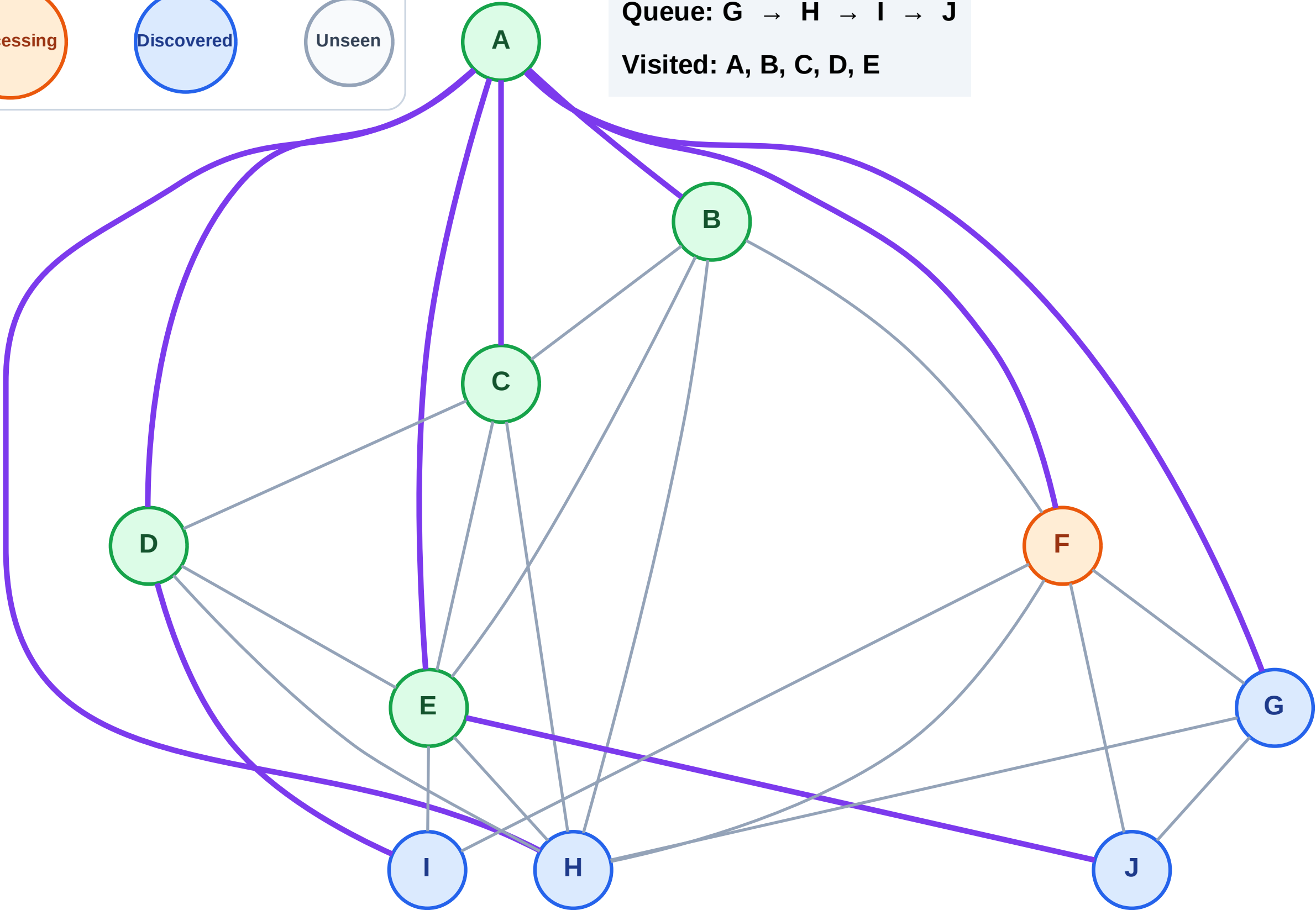
Processing

Discovered

Unseen

Queue: G → H → I → J

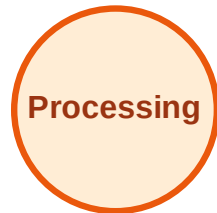
Visited: A, B, C, D, E



BFS Traversal

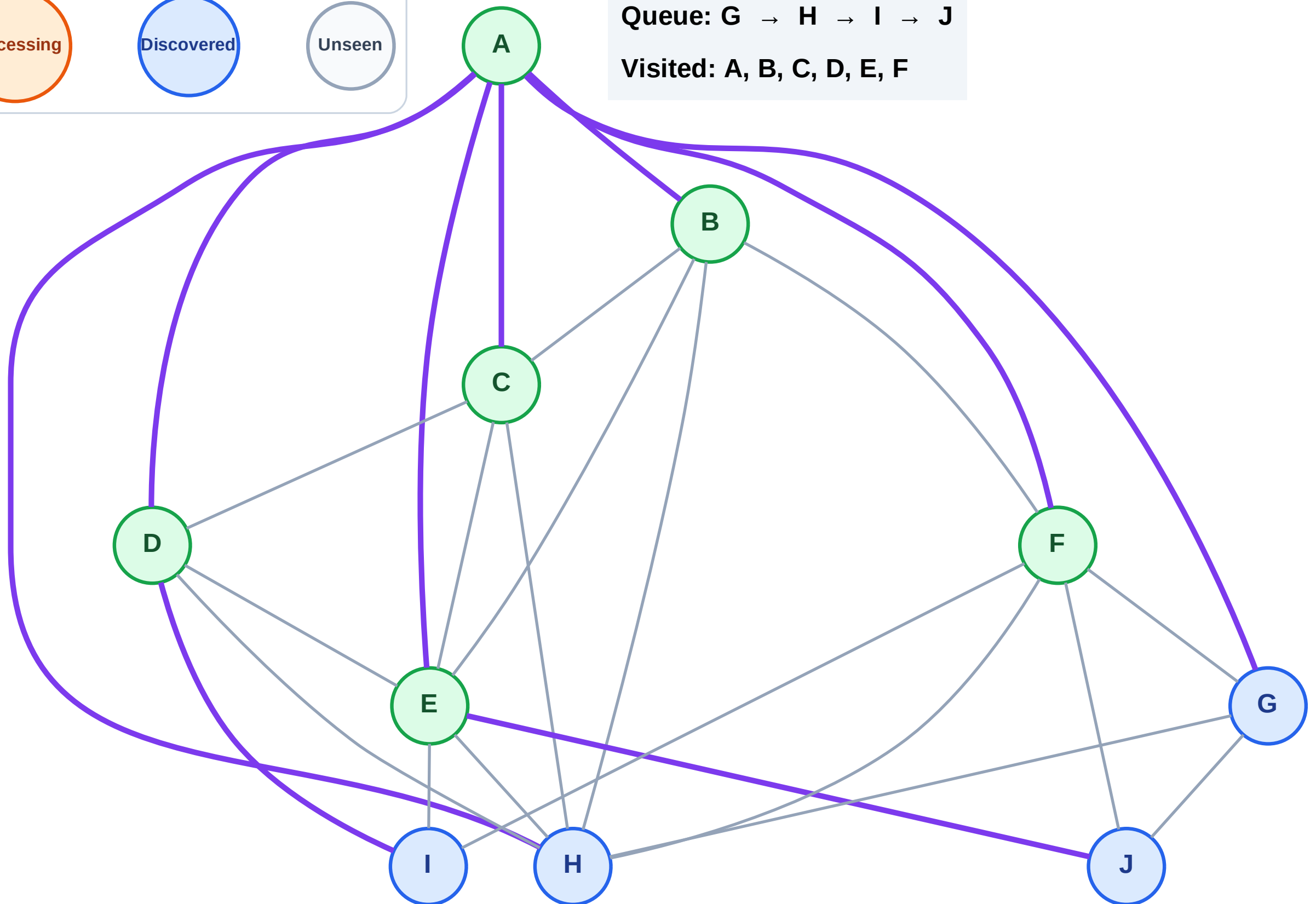
Step 13: No new neighbors from F

Legend



Queue: G → H → I → J

Visited: A, B, C, D, E, F



BFS Traversal

Step 14: Process node G

Legend

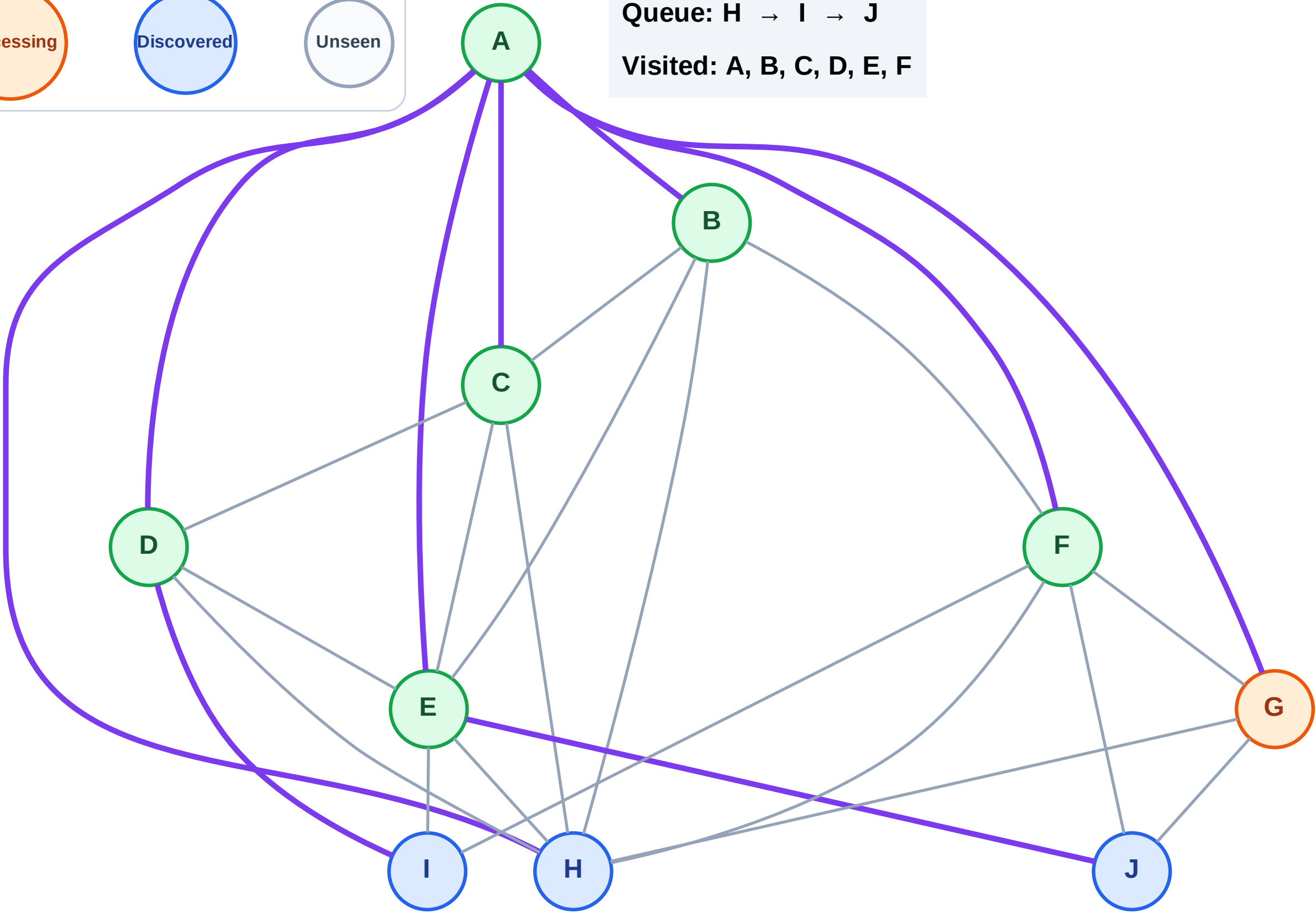
Complete

Processing

Discovered

Unseen

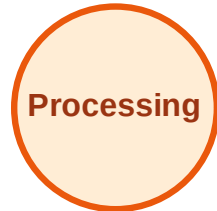
Queue: H → I → J
Visited: A, B, C, D, E, F



BFS Traversal

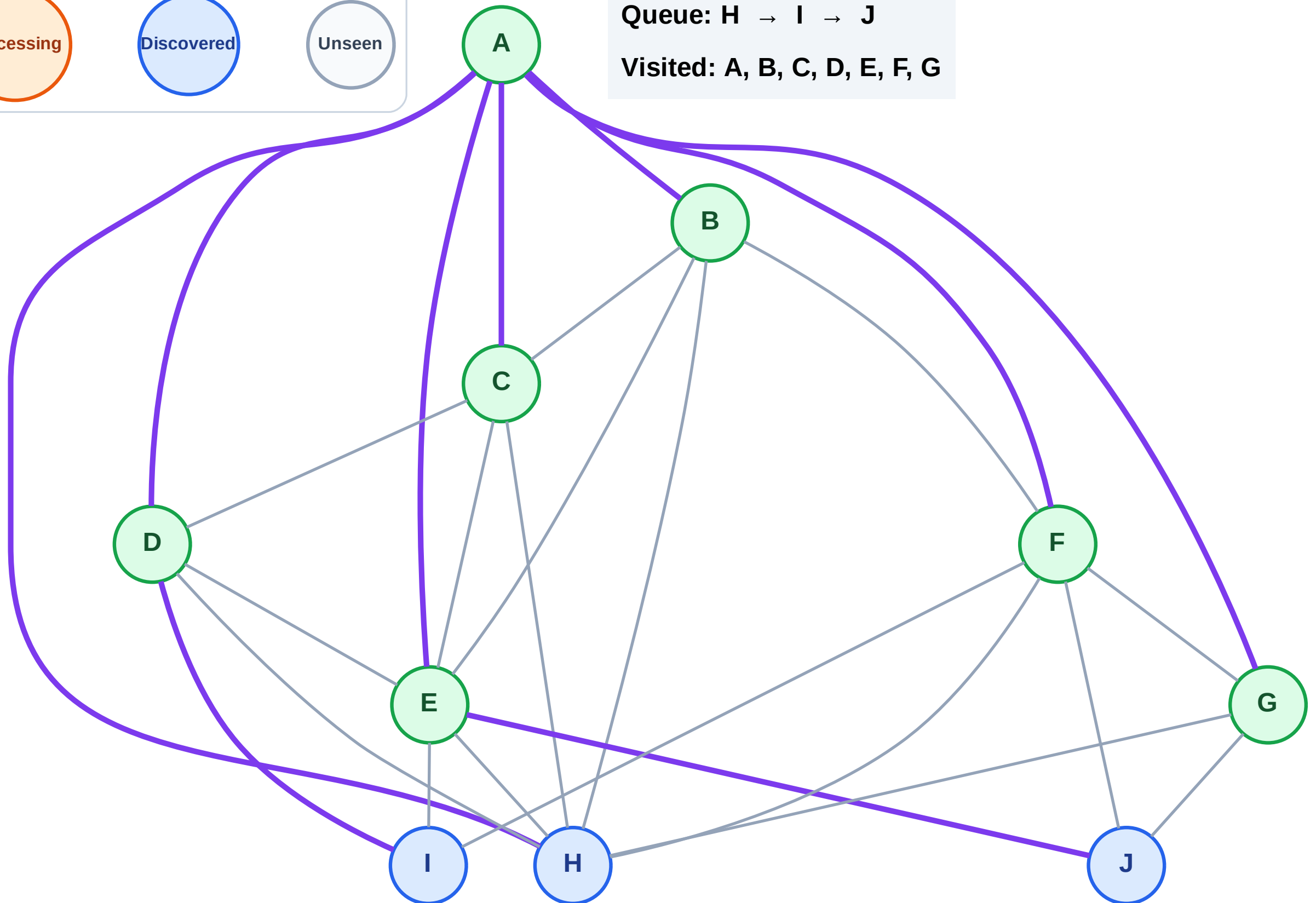
Step 15: No new neighbors from G

Legend



Queue: H → I → J

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node H

Legend

Complete

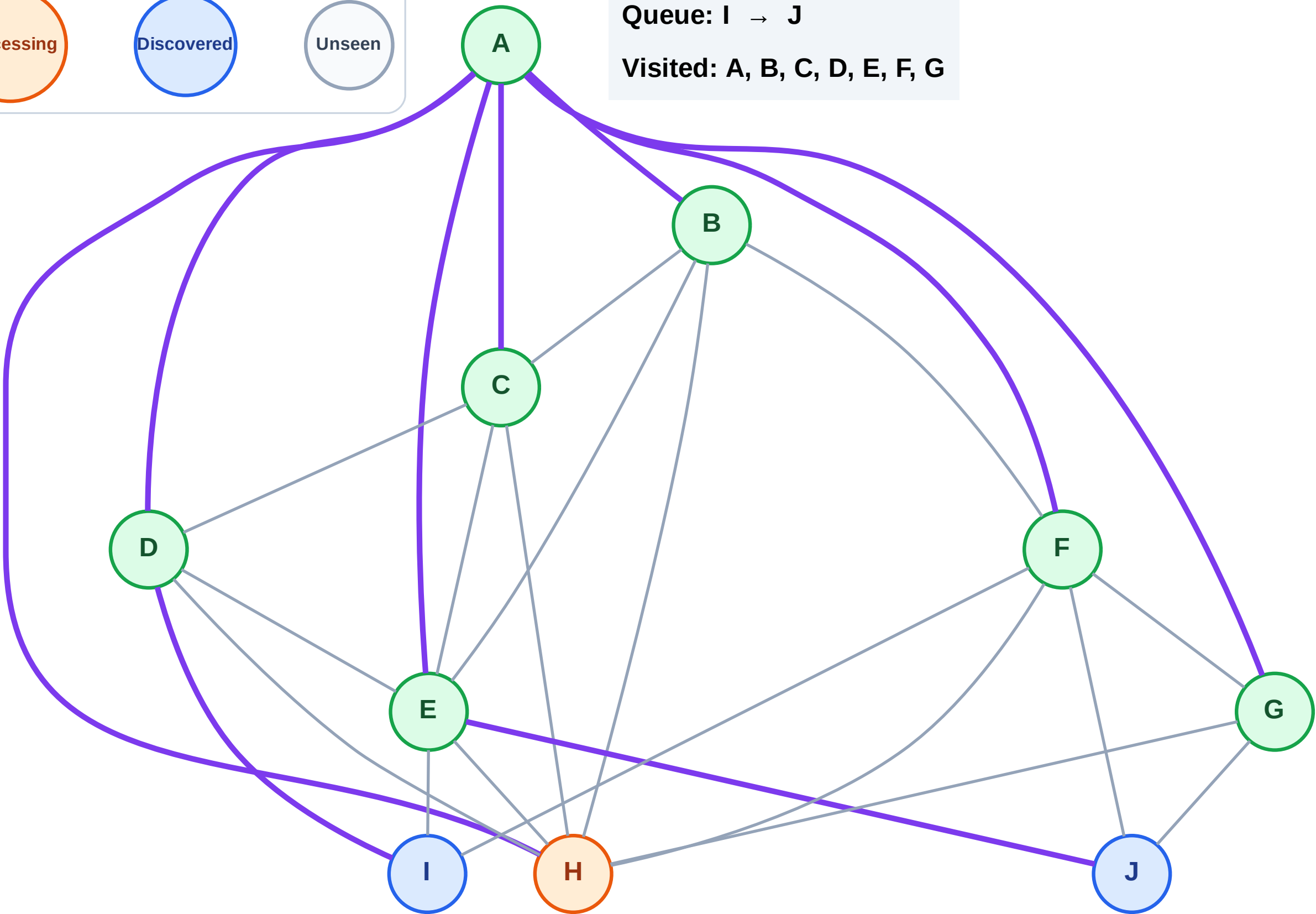
Processing

Discovered

Unseen

Queue: I → J

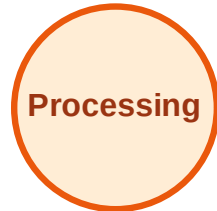
Visited: A, B, C, D, E, F, G



BFS Traversal

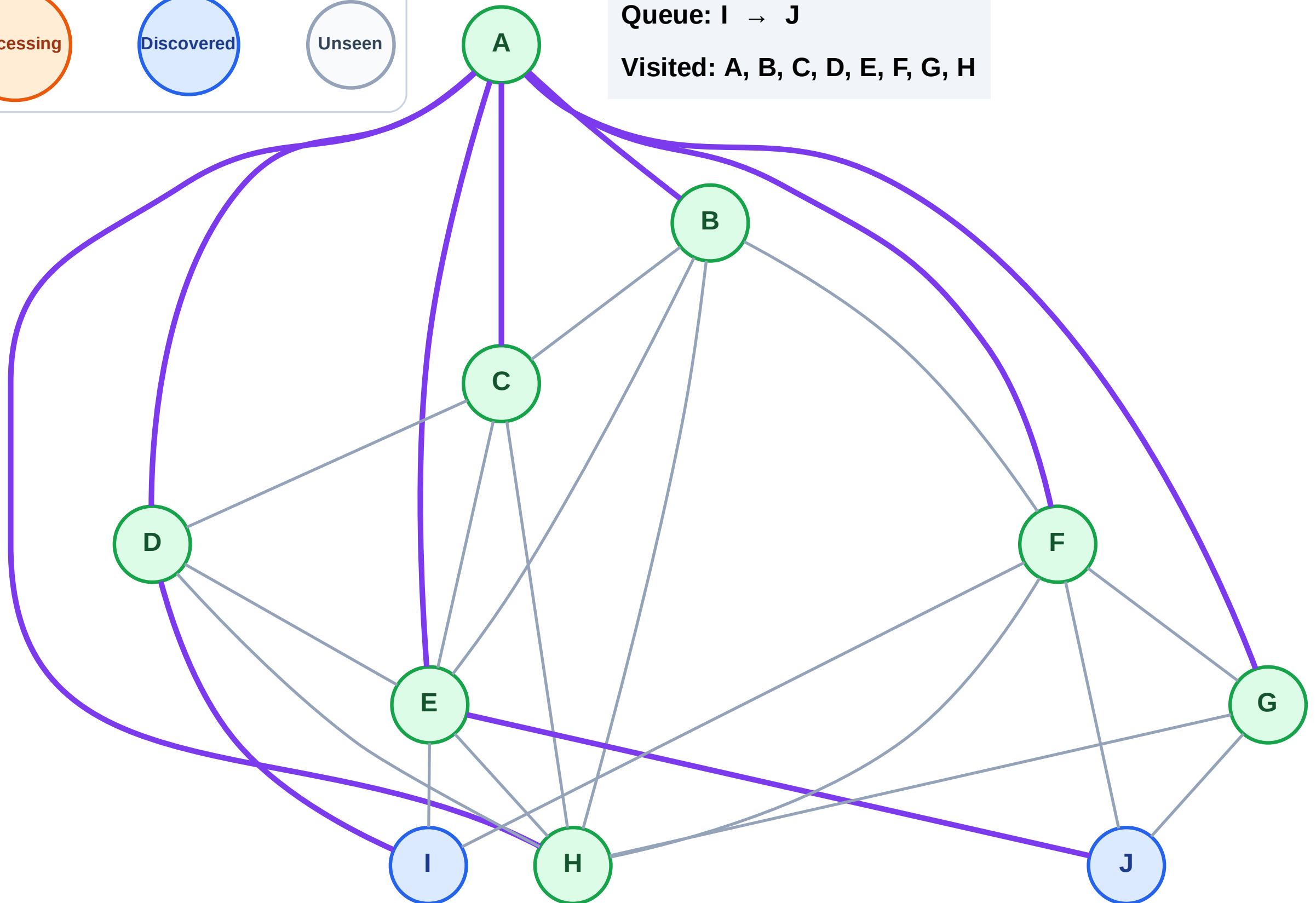
Step 17: No new neighbors from H

Legend



Queue: I → J

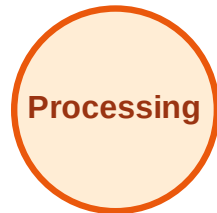
Visited: A, B, C, D, E, F, G, H



BFS Traversal

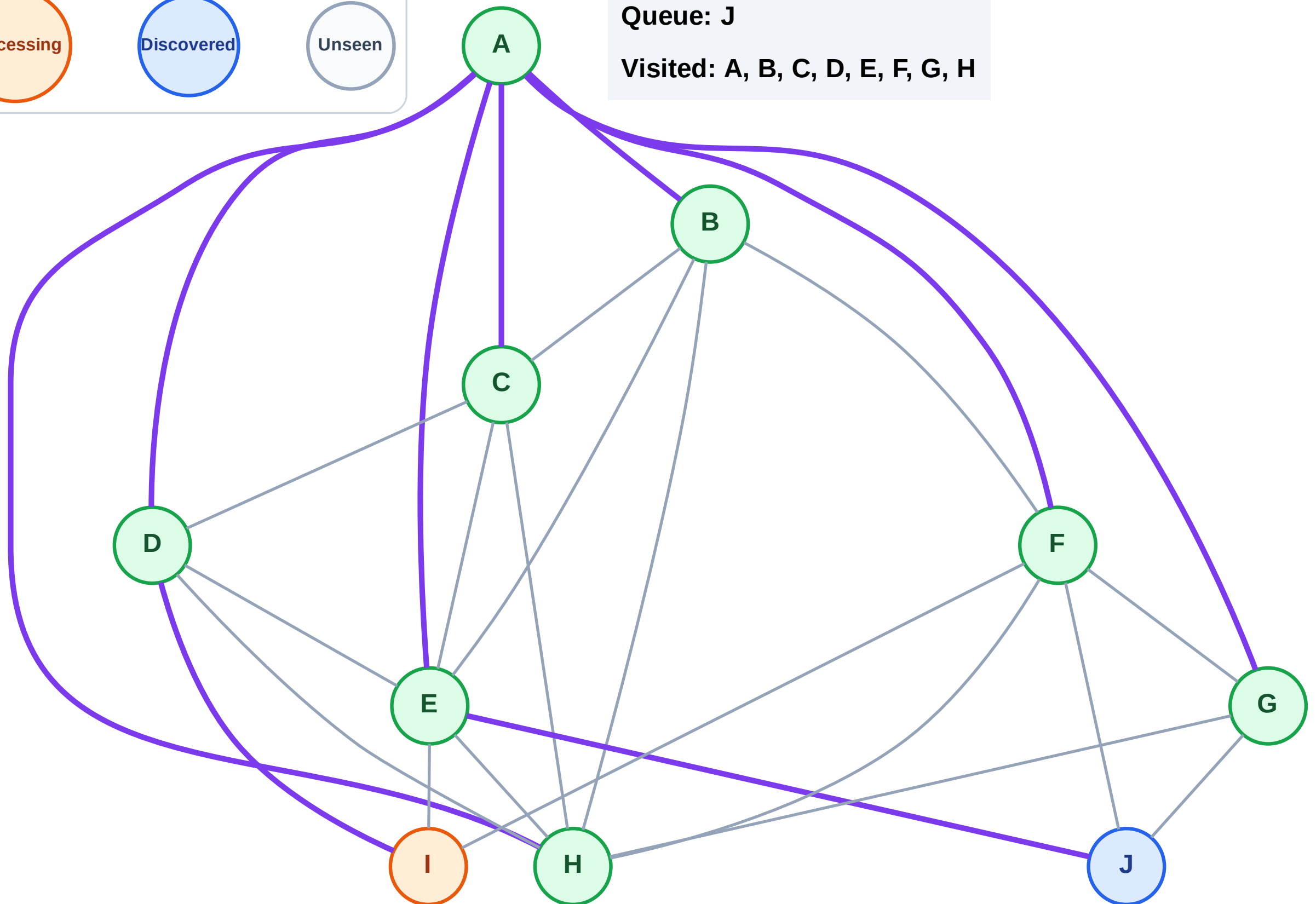
Step 18: Process node I

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H



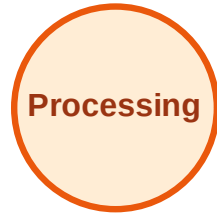
BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



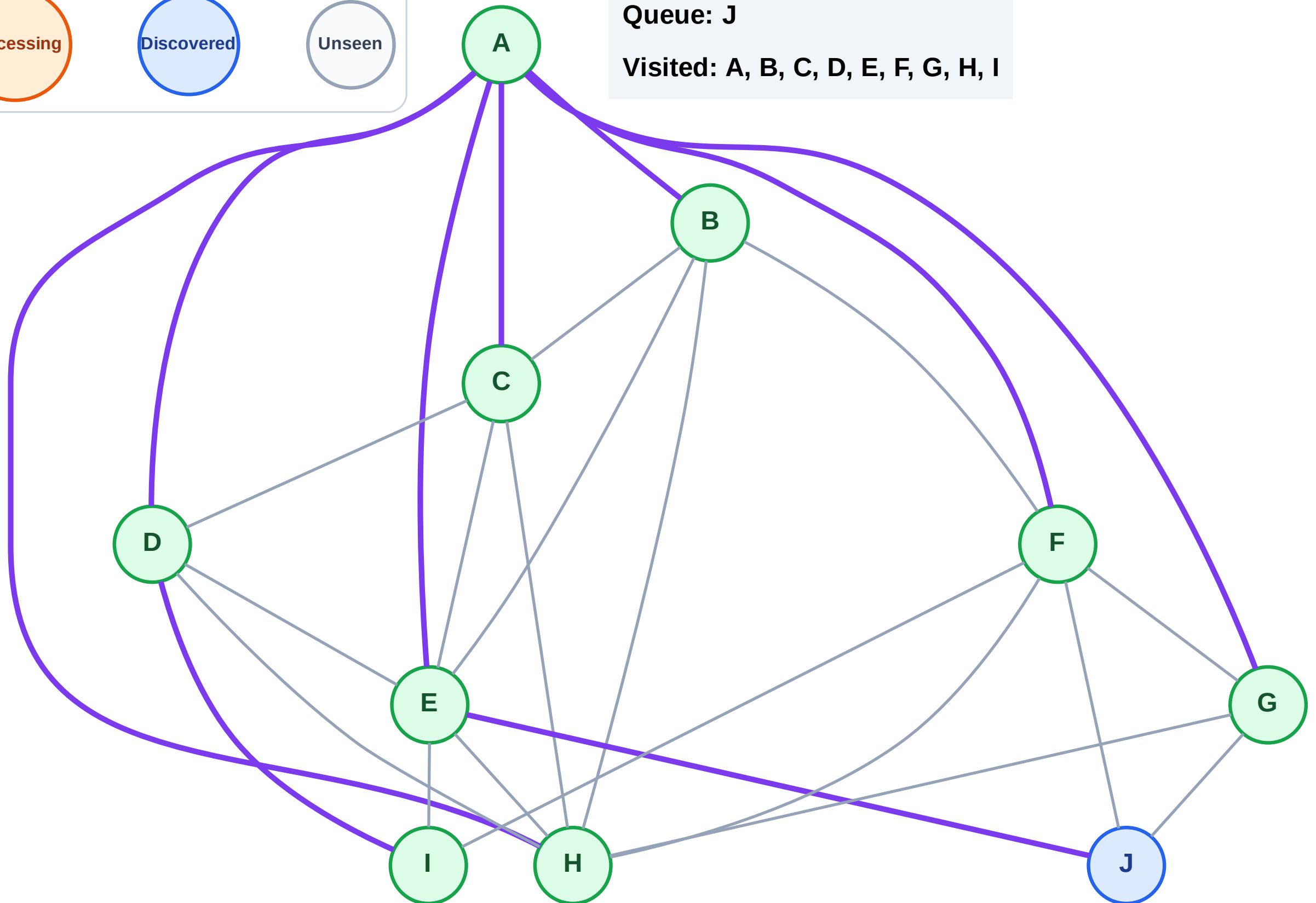
Discovered



Unseen

Queue: J

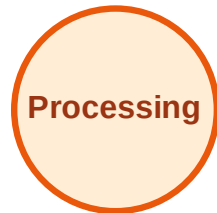
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

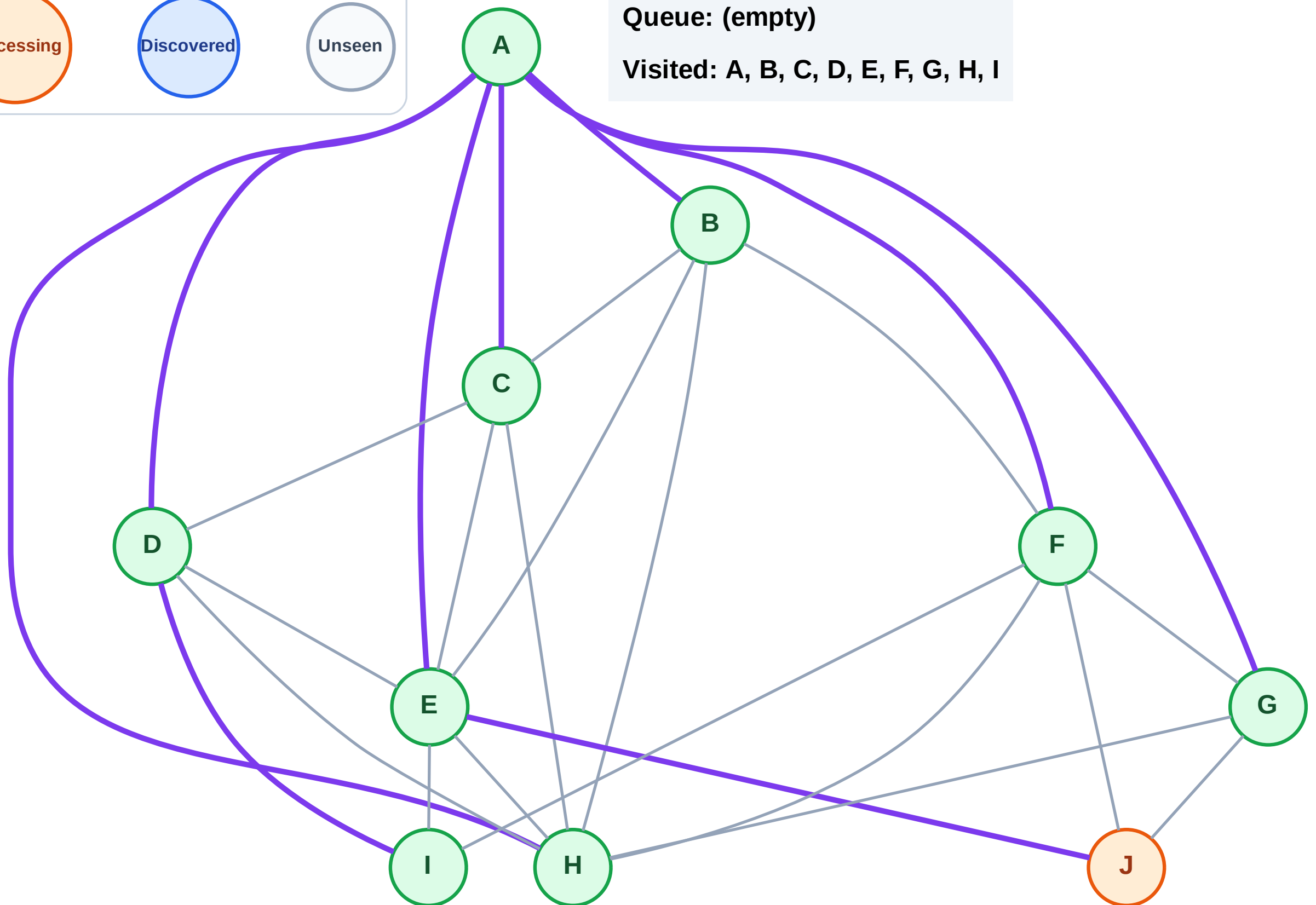
Step 20: Process node J

Legend



Queue: (empty)

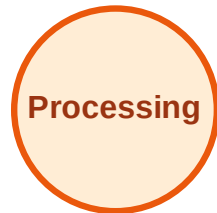
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

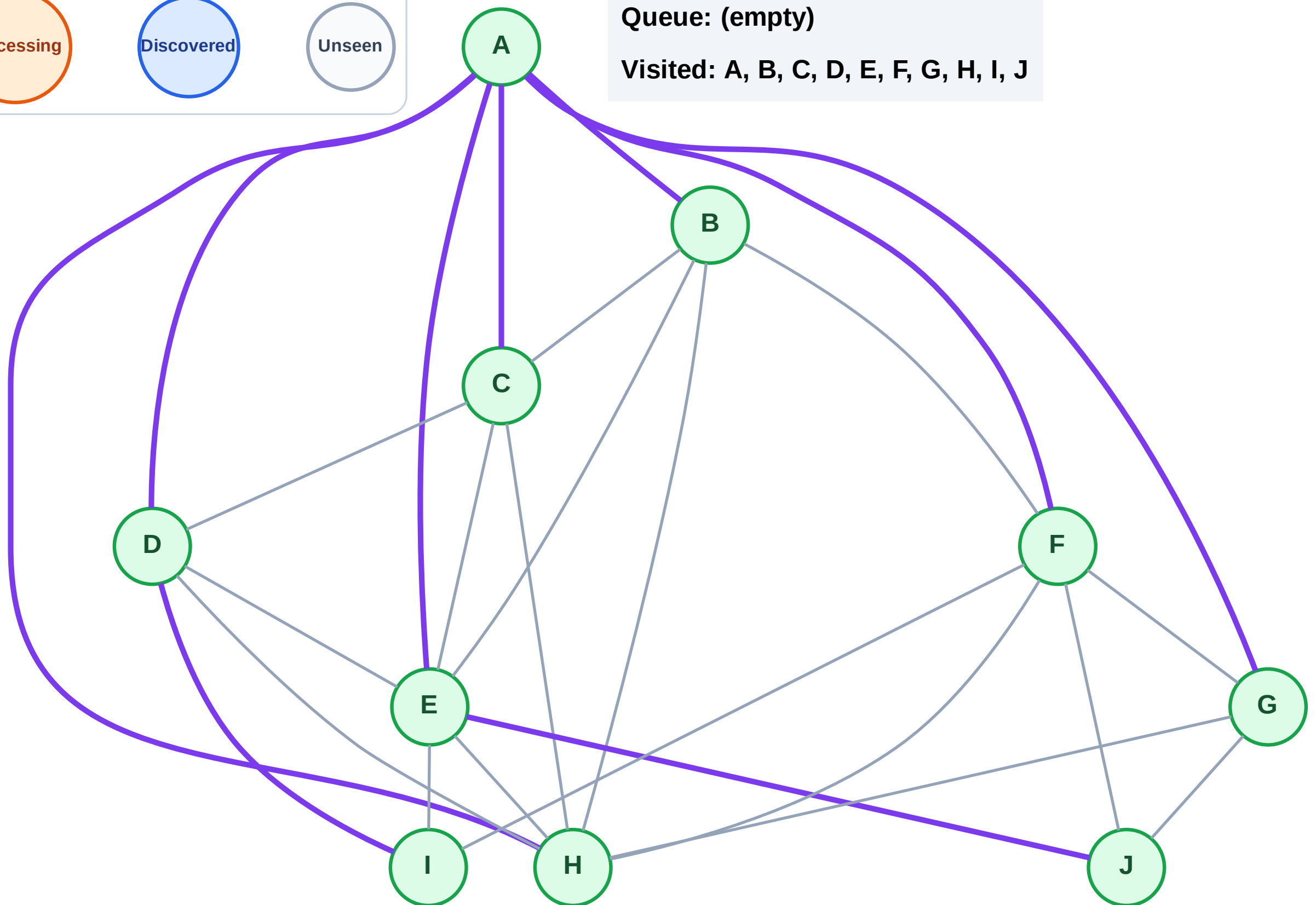
Step 21: No new neighbors from J

Legend



Queue: (empty)

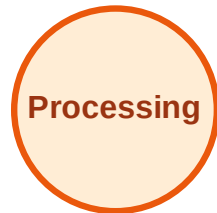
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

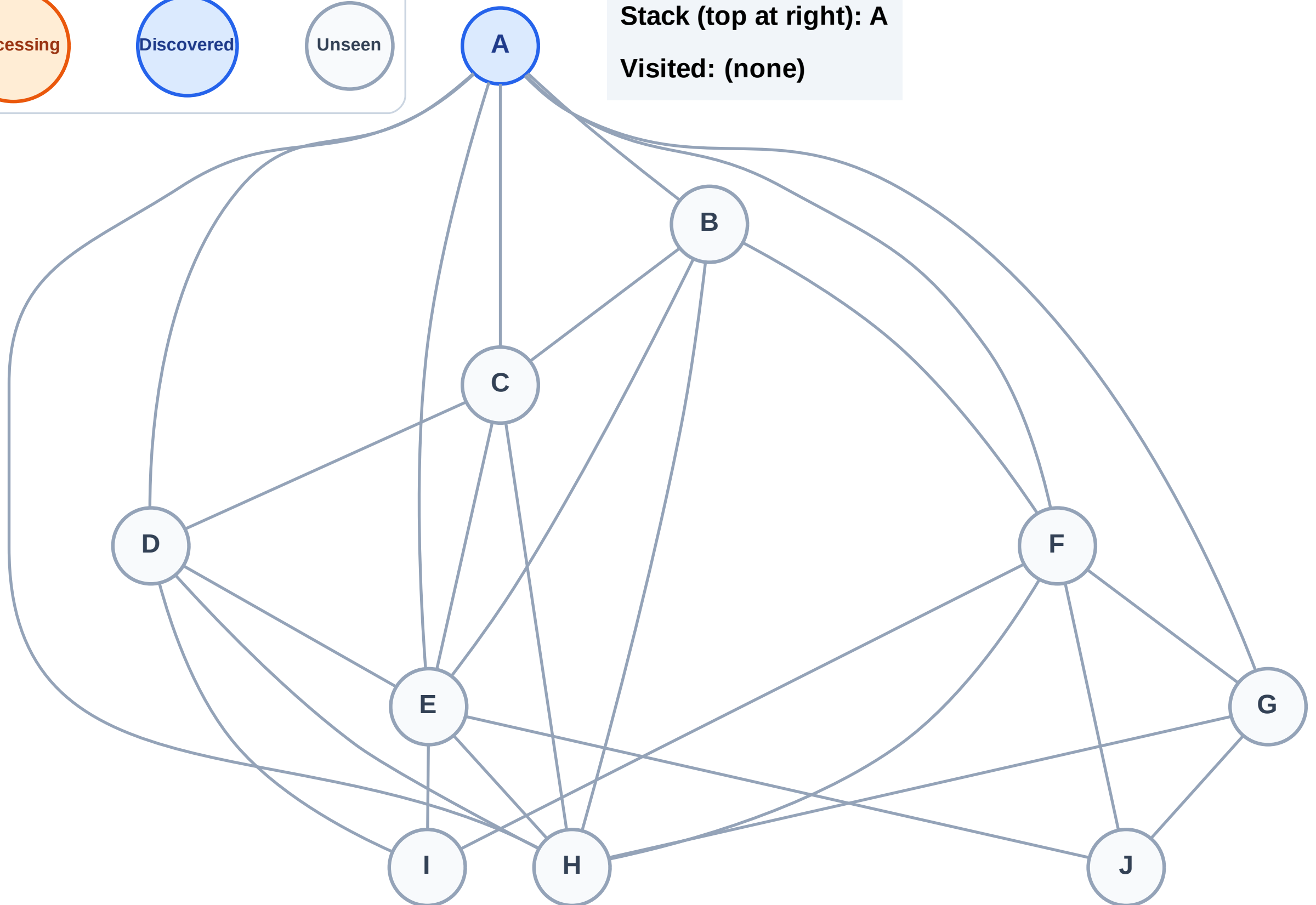
Step 1: Start at A

Legend



Stack (top at right): A

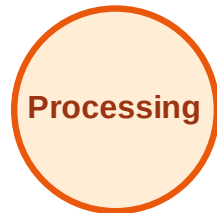
Visited: (none)



DFS Traversal

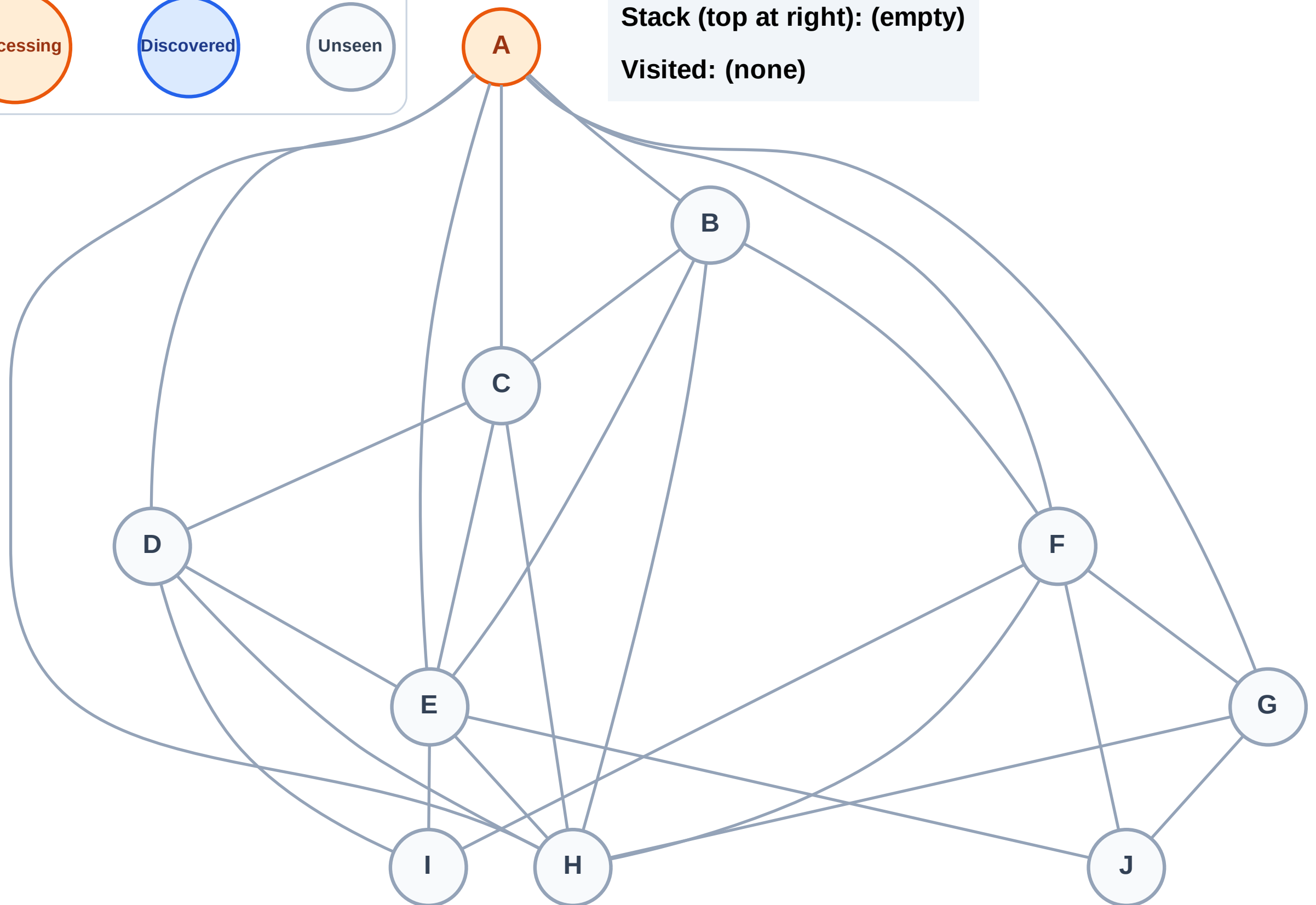
Step 2: Process node A

Legend



Stack (top at right): (empty)

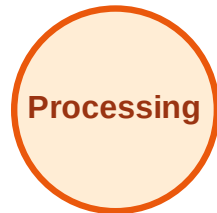
Visited: (none)



DFS Traversal

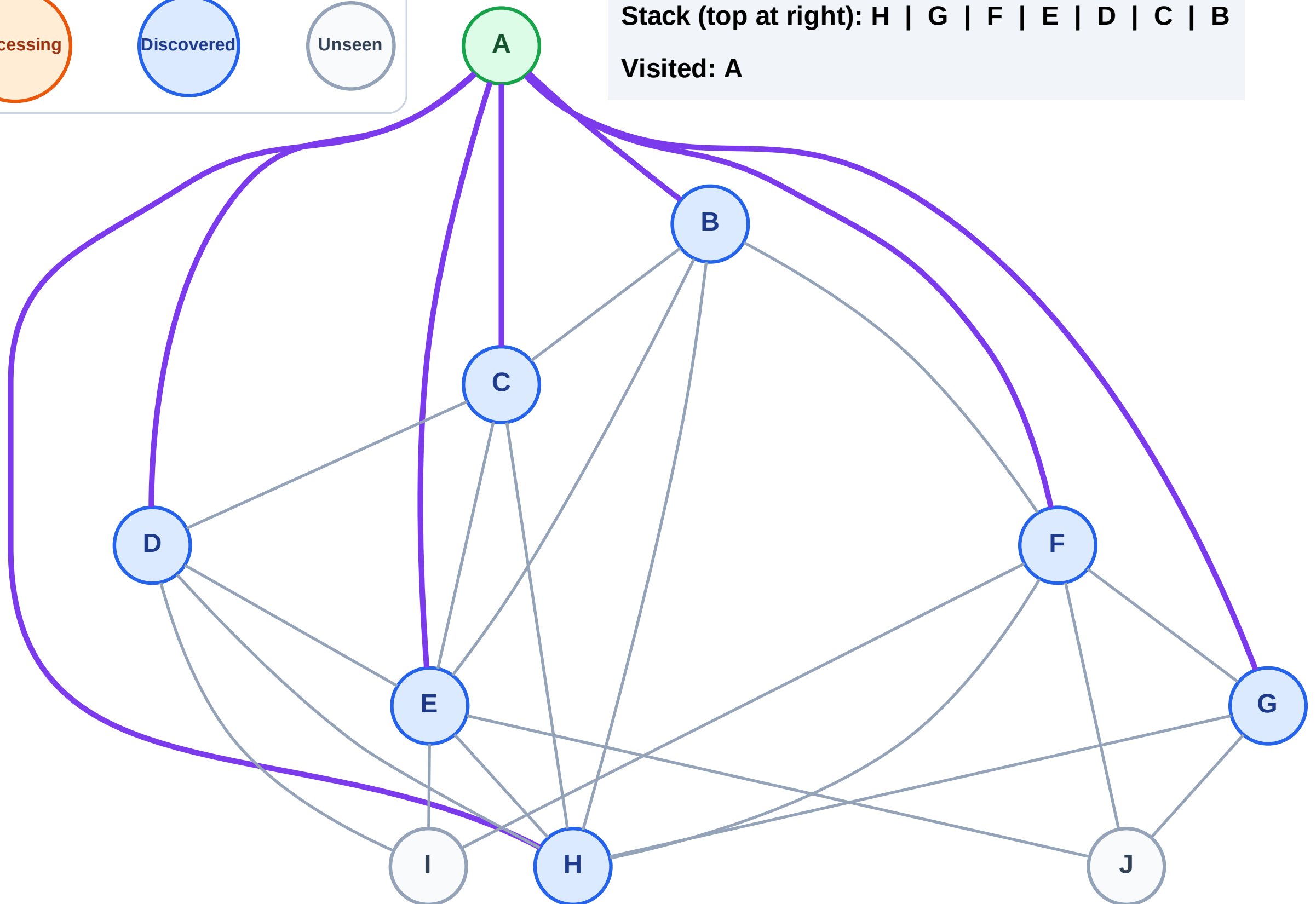
Step 3: Discover H, G, F, E, D, C, B from A

Legend



Stack (top at right): H | G | F | E | D | C | B

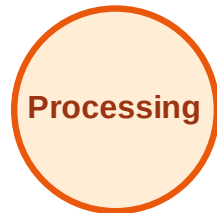
Visited: A



DFS Traversal

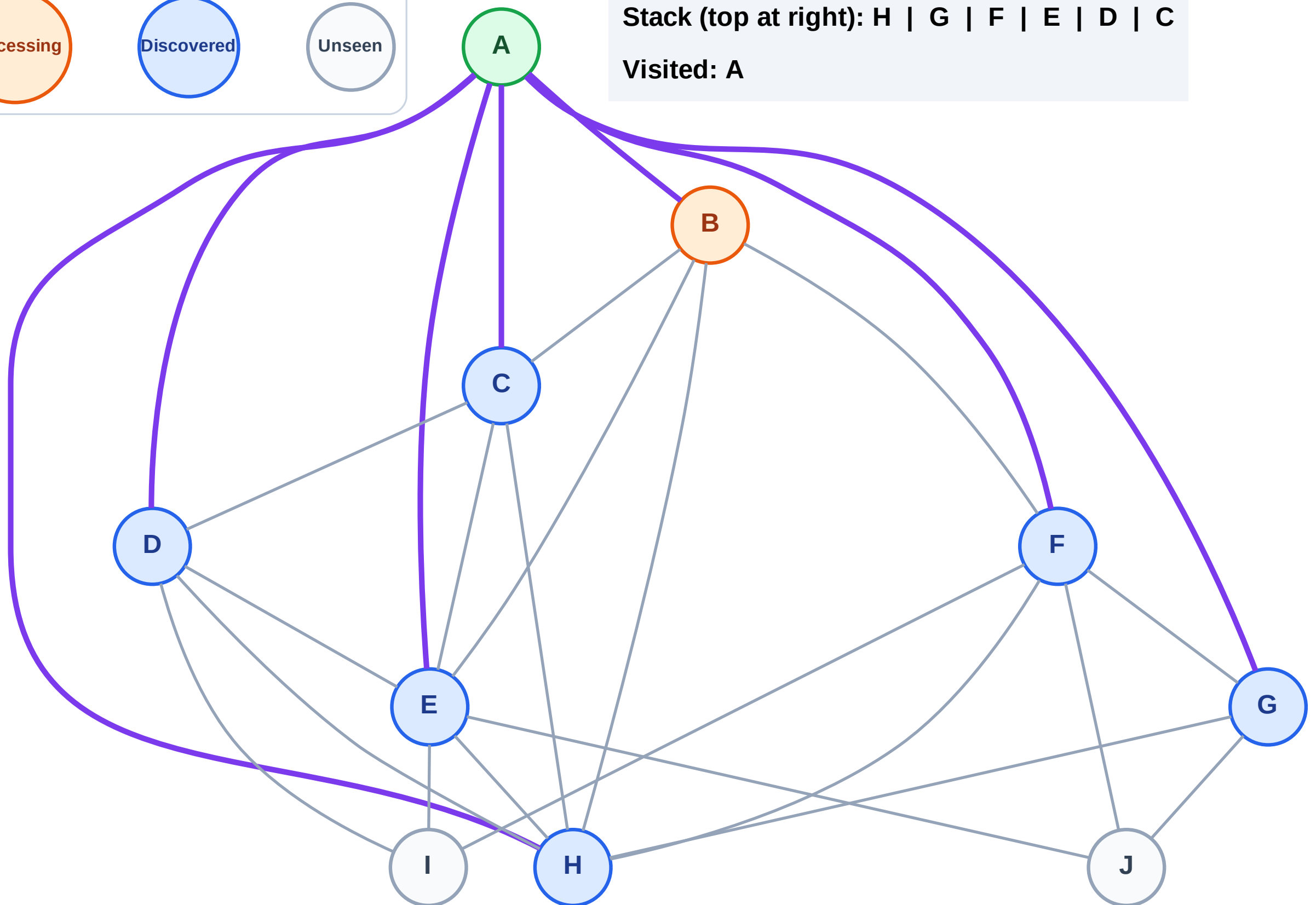
Step 4: Process node B

Legend



Stack (top at right): H | G | F | E | D | C

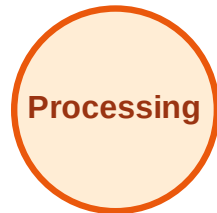
Visited: A



DFS Traversal

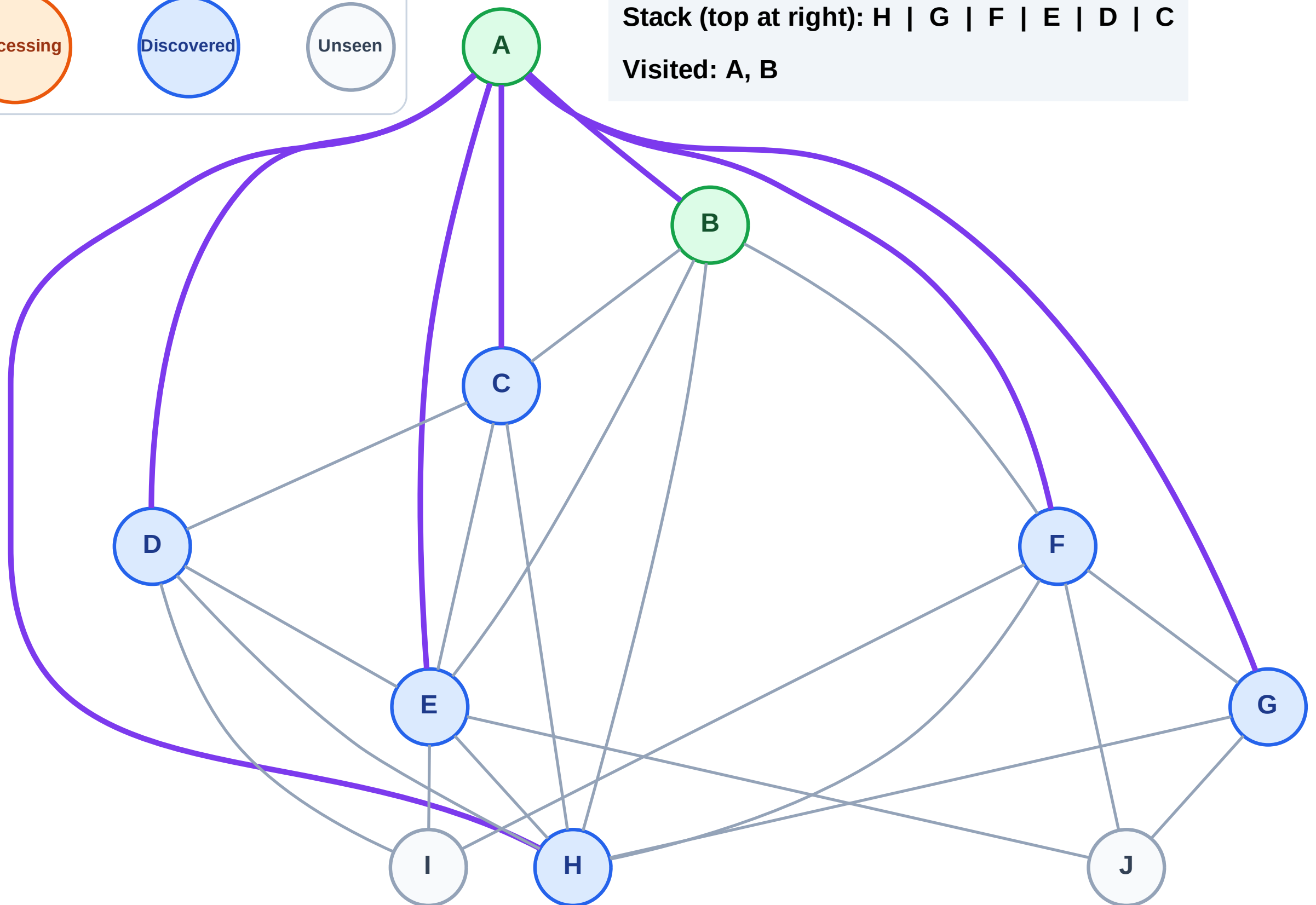
Step 5: No new neighbors from B

Legend



Stack (top at right): H | G | F | E | D | C

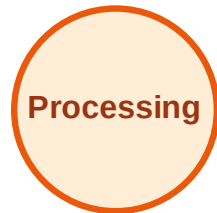
Visited: A, B



DFS Traversal

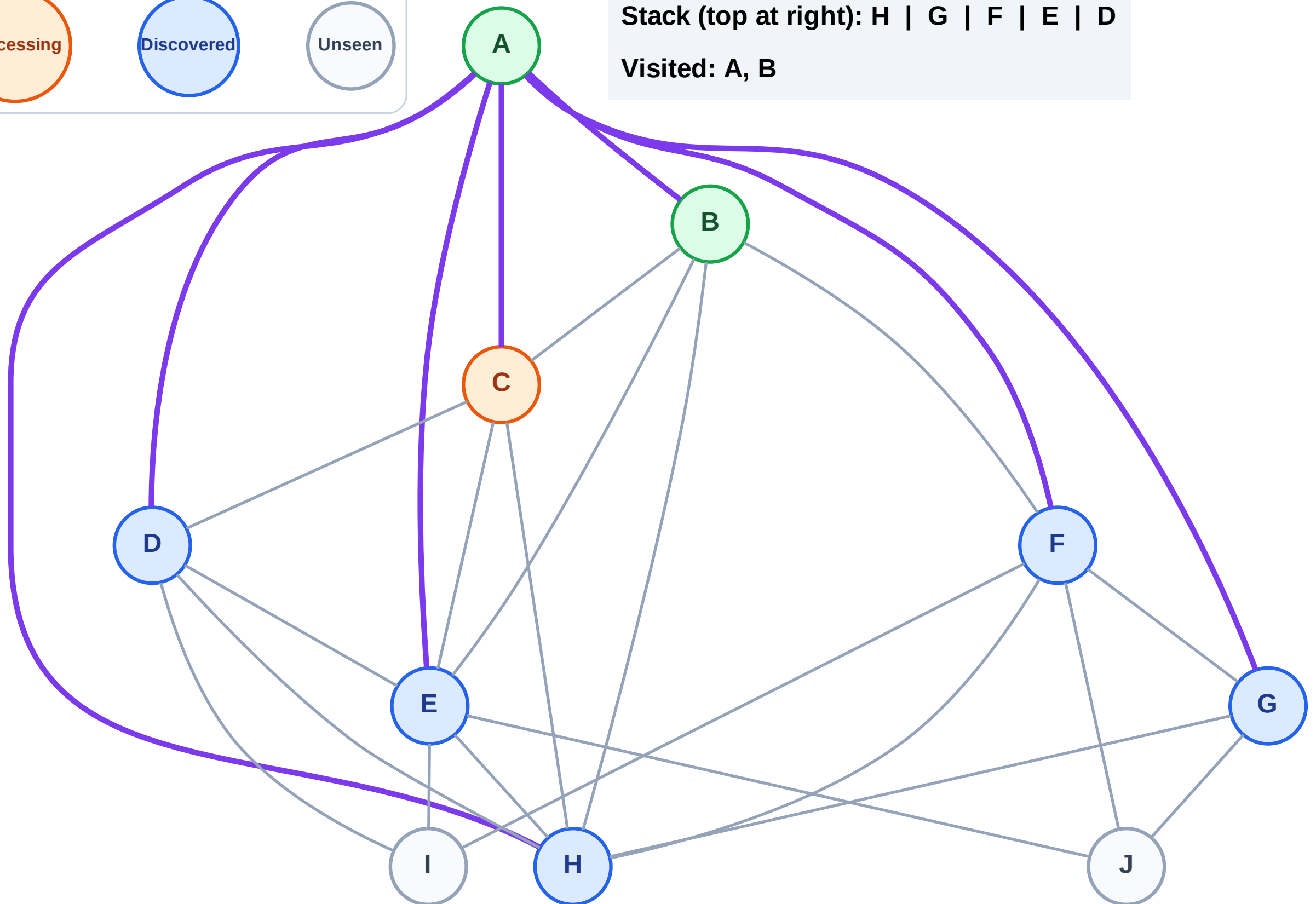
Step 6: Process node C

Legend



Stack (top at right): H | G | F | E | D

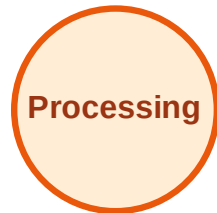
Visited: A, B



DFS Traversal

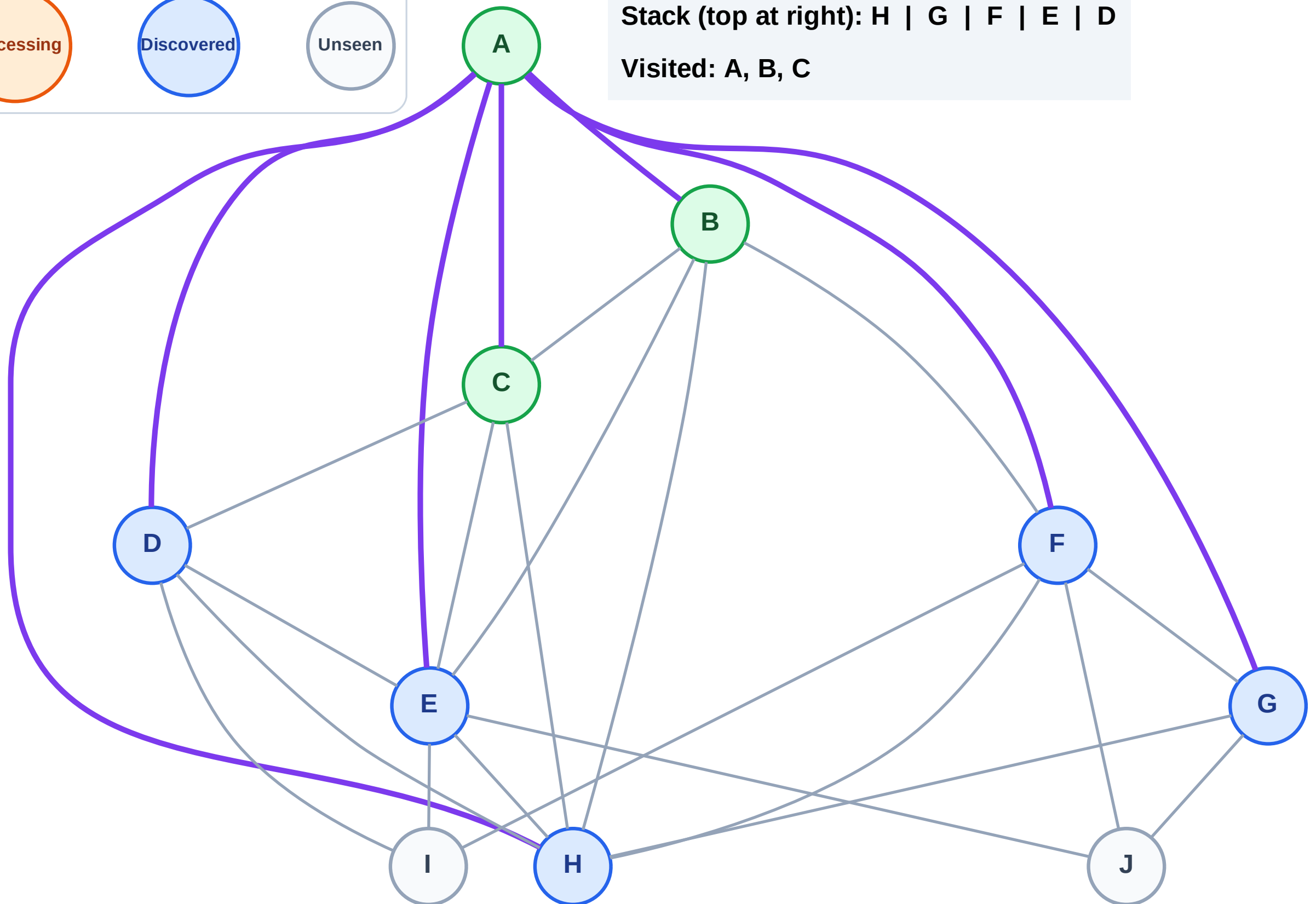
Step 7: No new neighbors from C

Legend



Stack (top at right): H | G | F | E | D

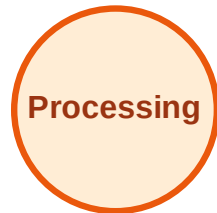
Visited: A, B, C



DFS Traversal

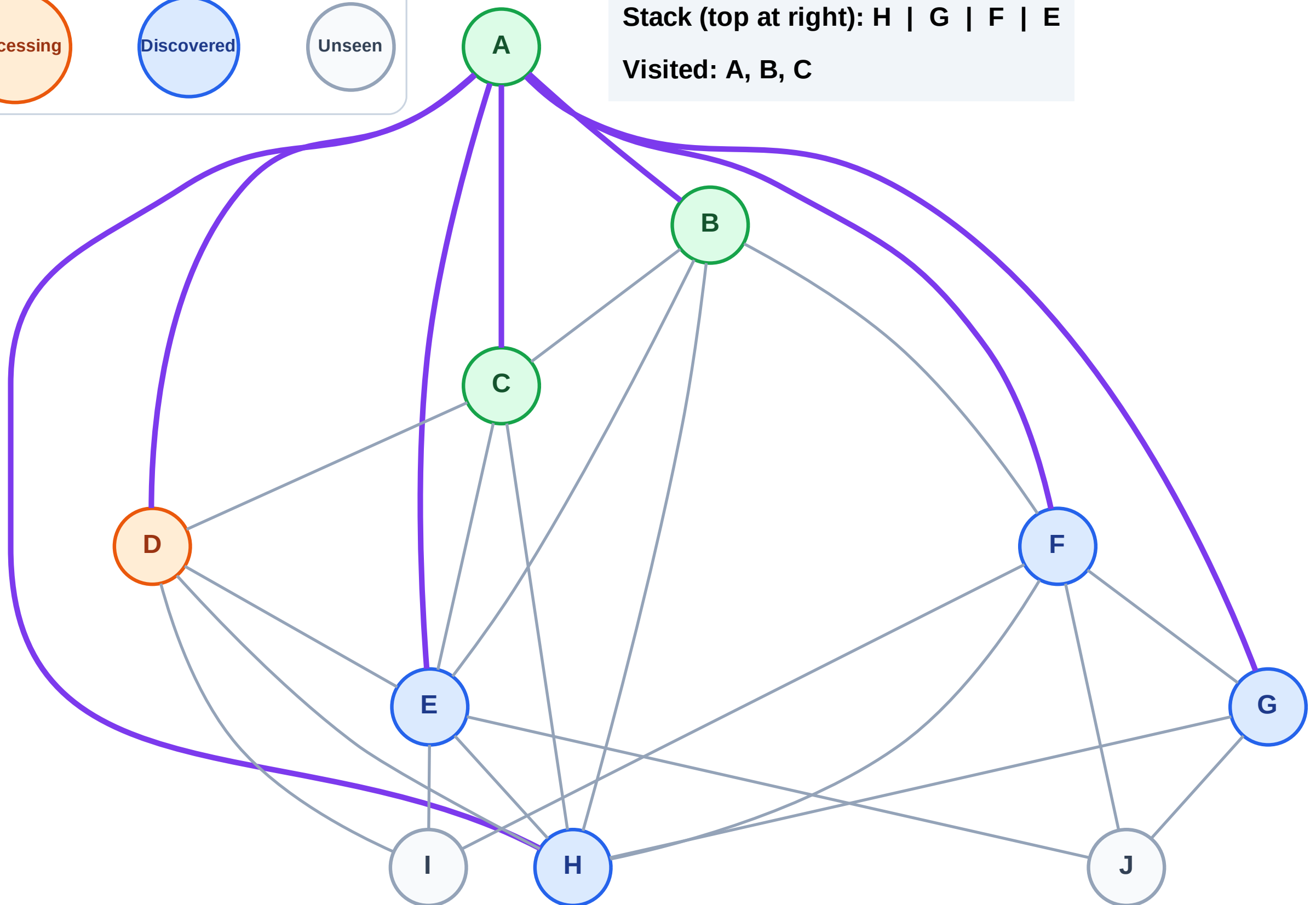
Step 8: Process node D

Legend



Stack (top at right): H | G | F | E

Visited: A, B, C



DFS Traversal

Step 9: Discover I from D

Legend

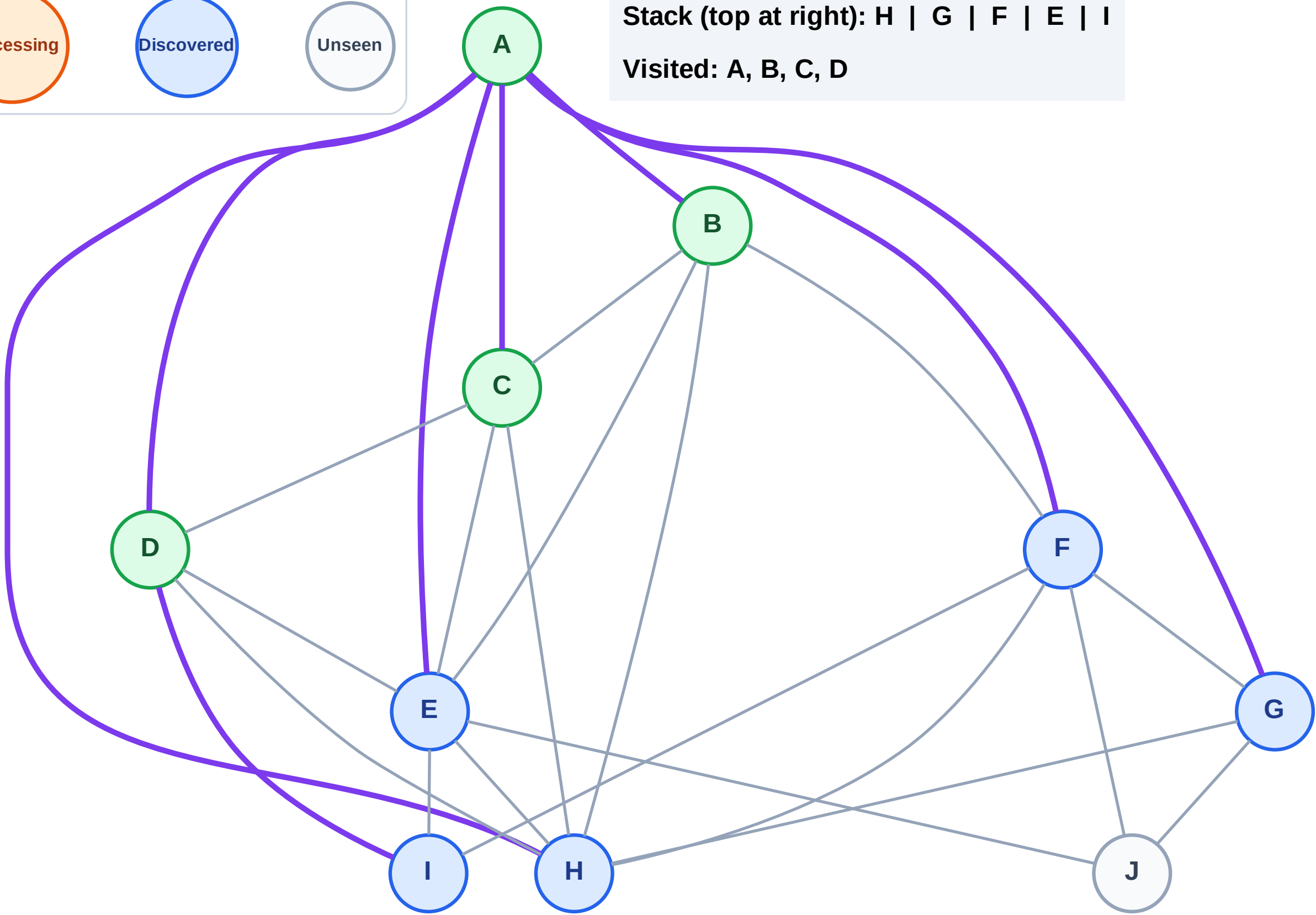
Complete

Processing

Discovered

Unseen

Stack (top at right): H | G | F | E | I
Visited: A, B, C, D



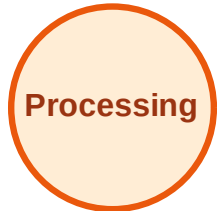
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



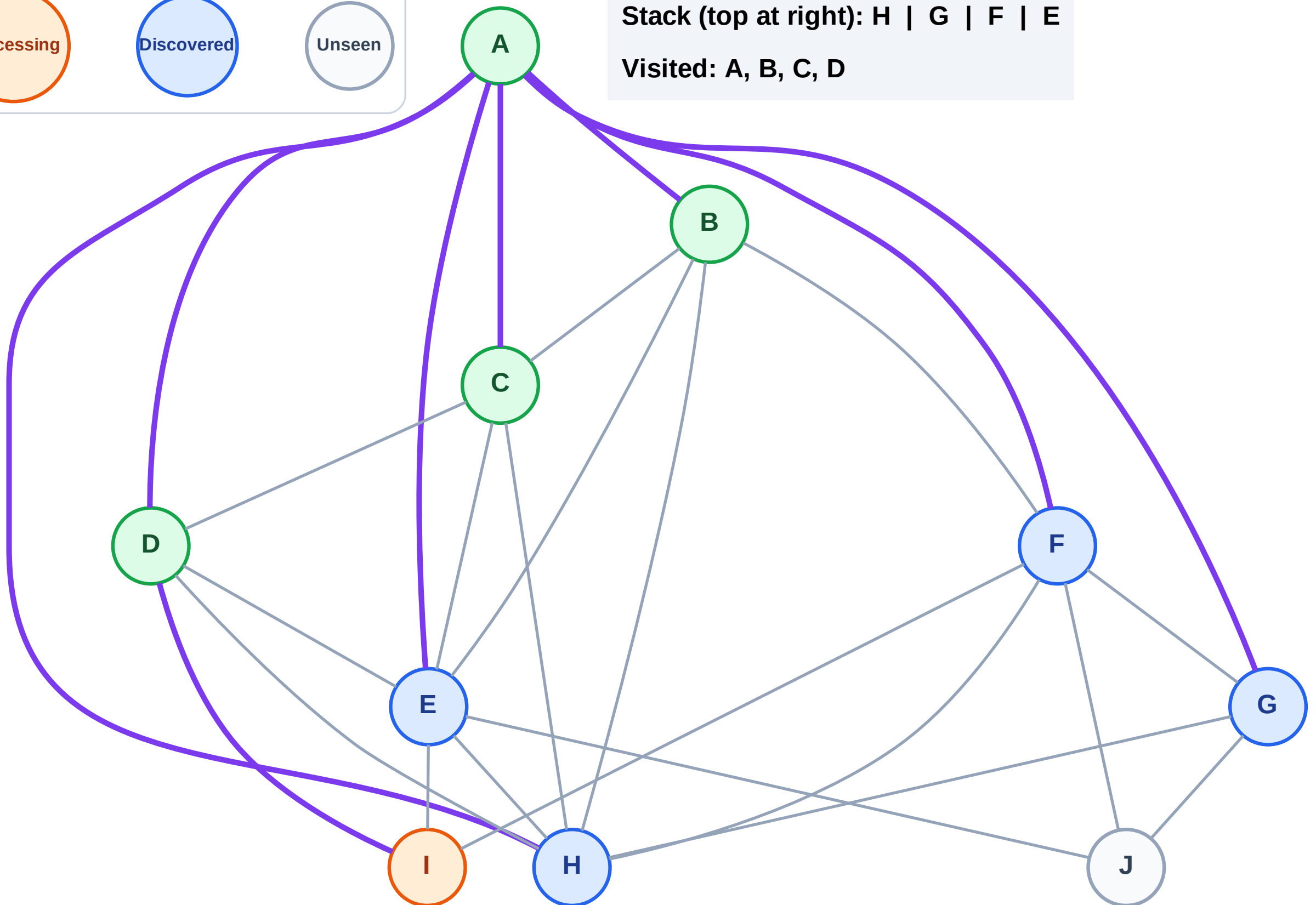
Discovered



Unseen

Stack (top at right): H | G | F | E

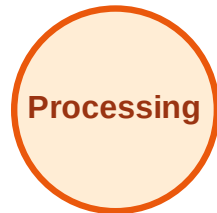
Visited: A, B, C, D



DFS Traversal

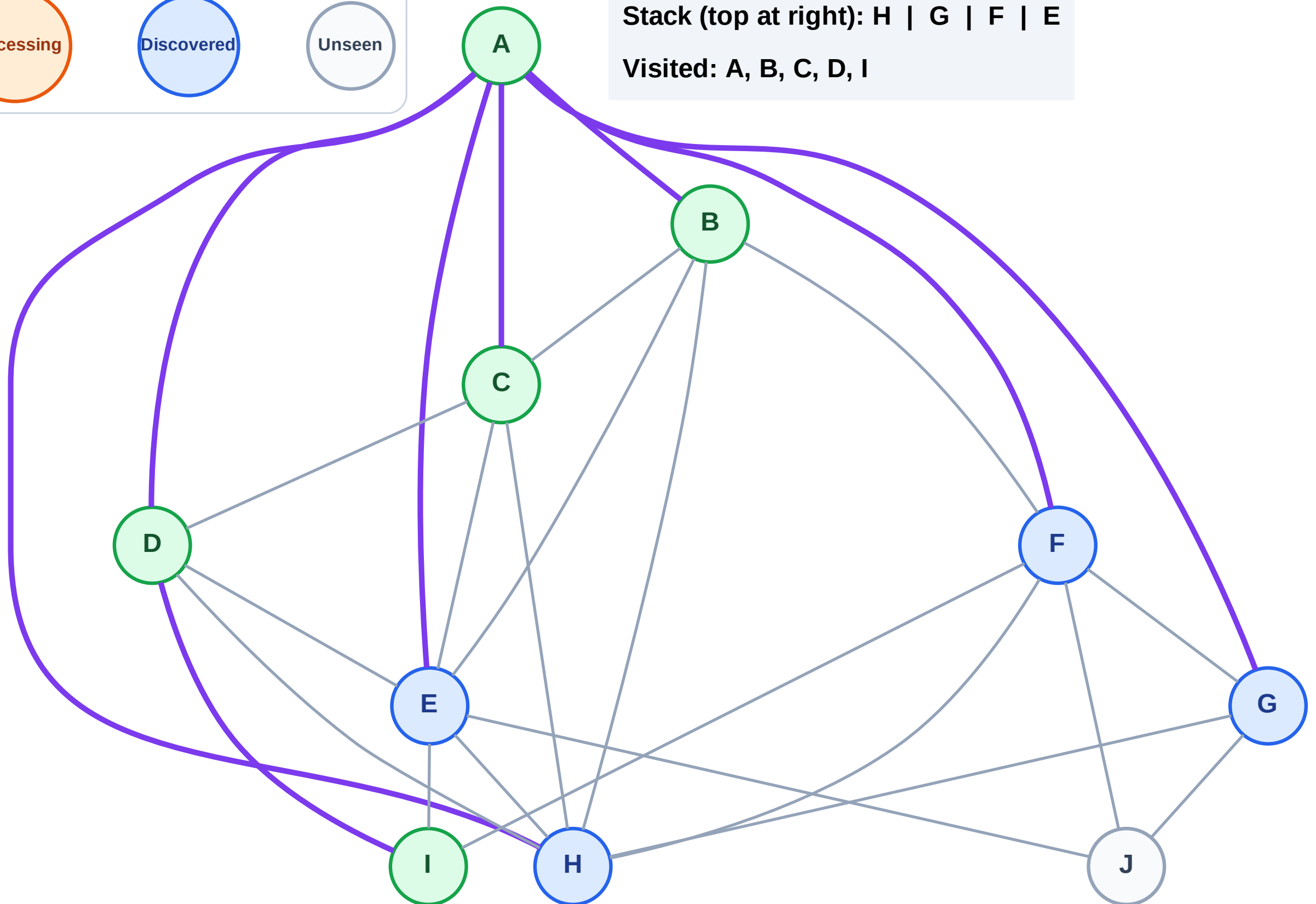
Step 11: No new neighbors from I

Legend



Stack (top at right): H | G | F | E

Visited: A, B, C, D, I



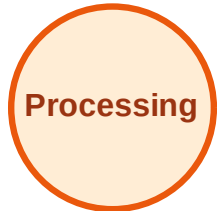
DFS Traversal

Step 12: Process node E

Legend



Complete



Processing



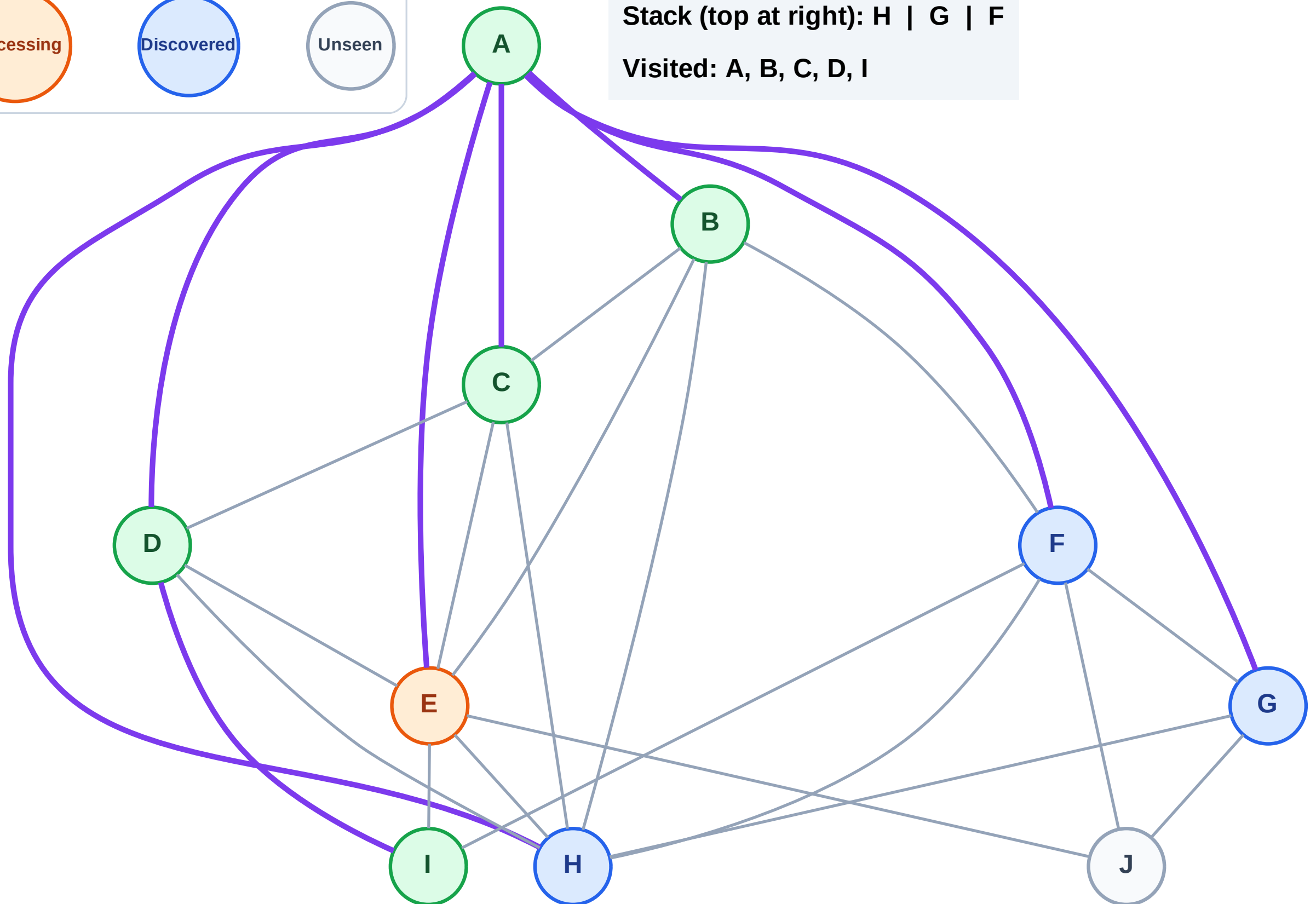
Discovered



Unseen

Stack (top at right): H | G | F

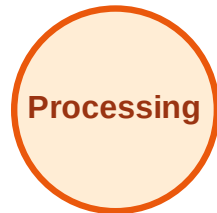
Visited: A, B, C, D, I



DFS Traversal

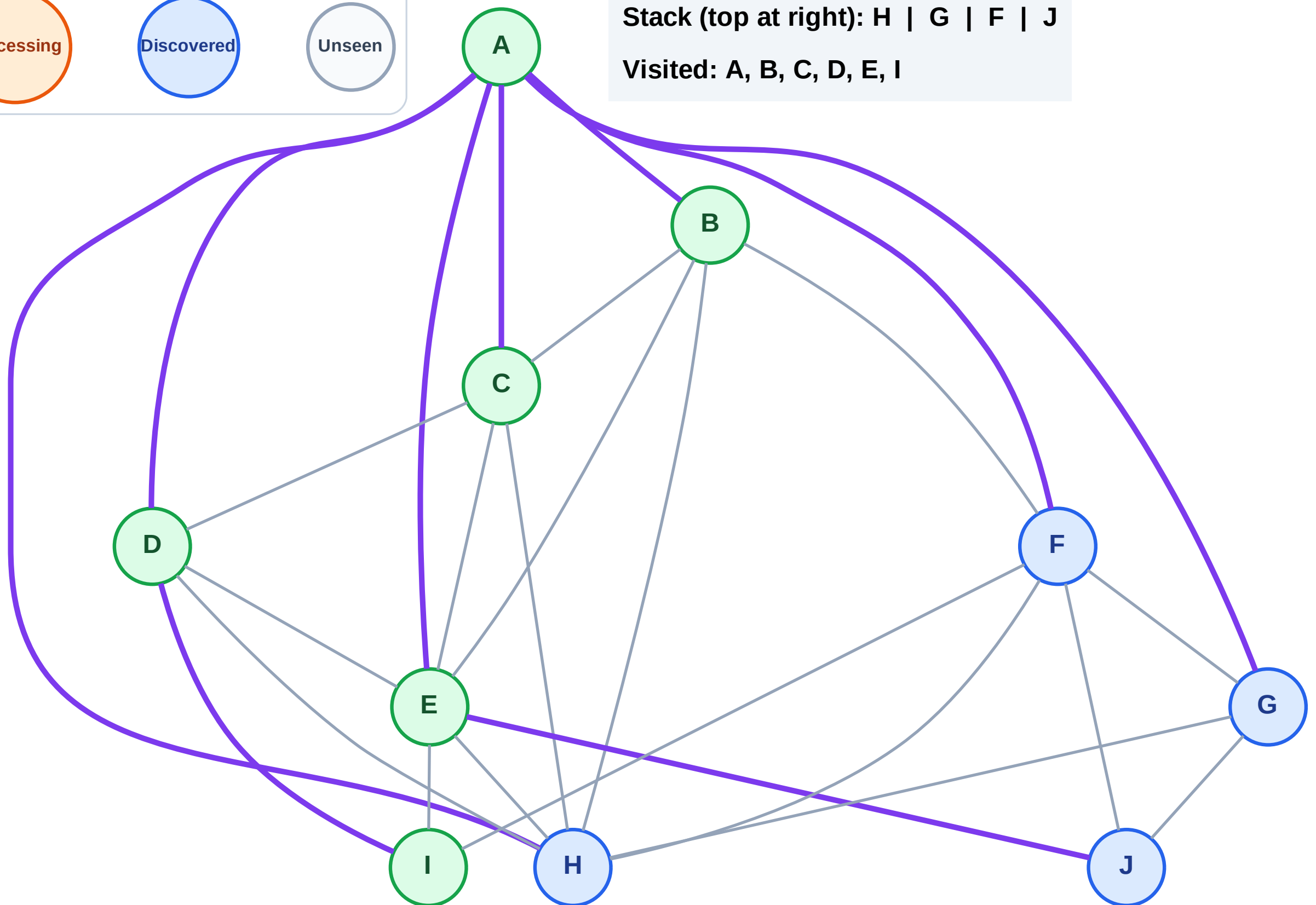
Step 13: Discover J from E

Legend



Stack (top at right): H | G | F | J

Visited: A, B, C, D, E, I



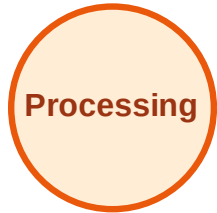
DFS Traversal

Step 14: Process node J

Legend



Complete



Processing



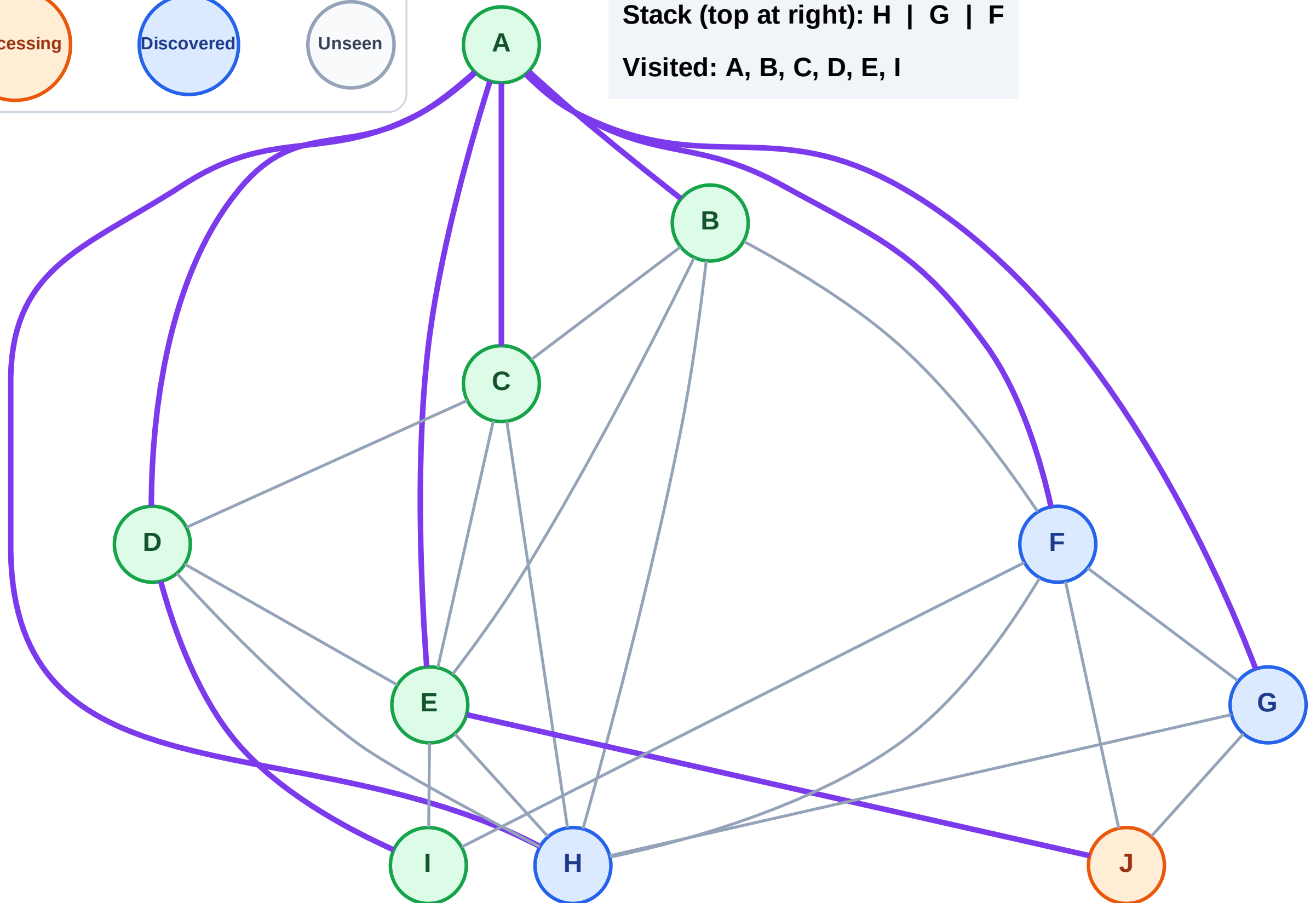
Discovered



Unseen

Stack (top at right): H | G | F

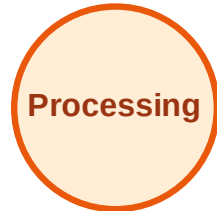
Visited: A, B, C, D, E, I



DFS Traversal

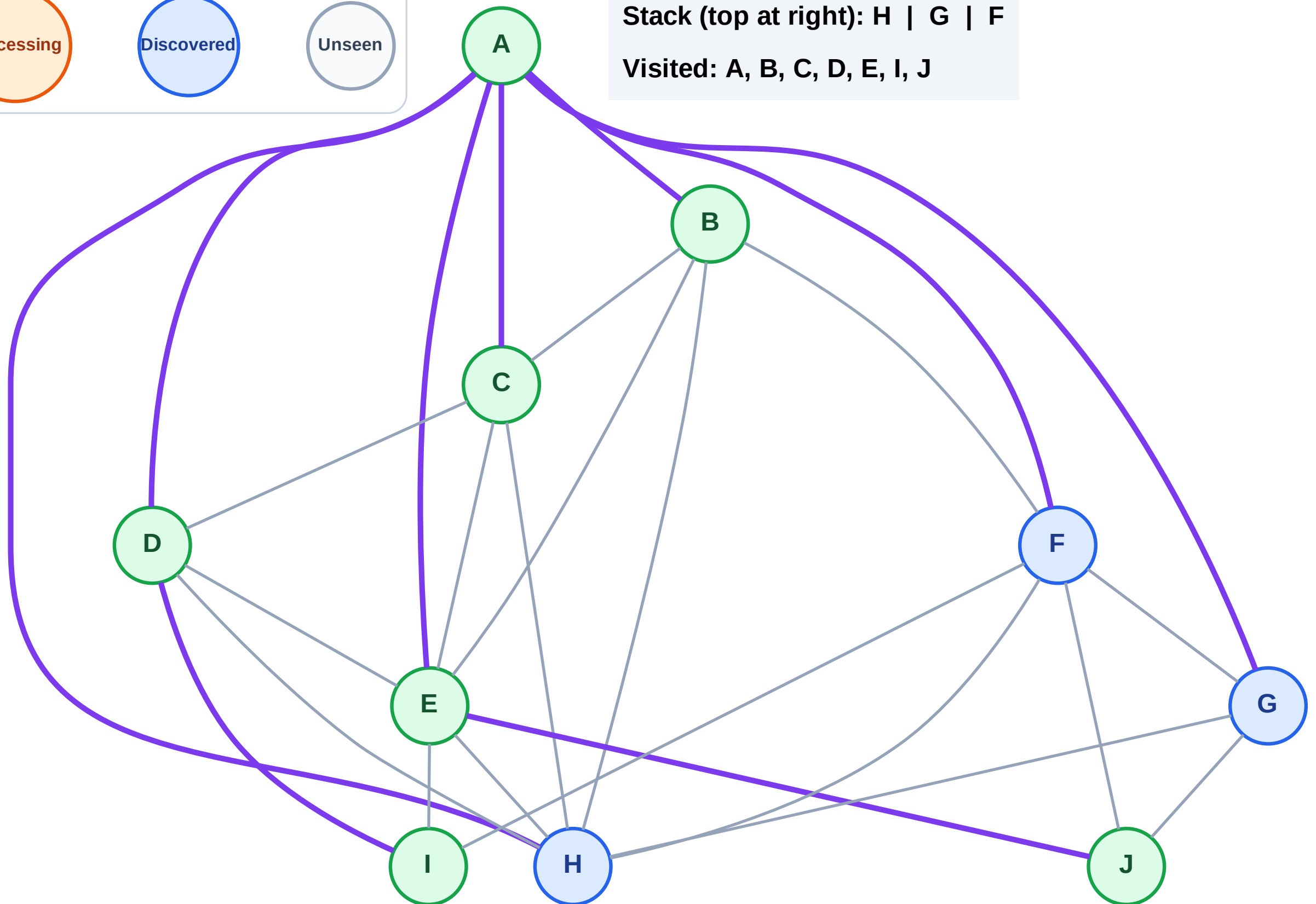
Step 15: No new neighbors from J

Legend



Stack (top at right): H | G | F

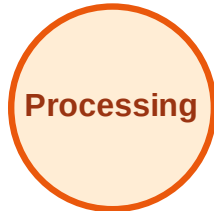
Visited: A, B, C, D, E, I, J



DFS Traversal

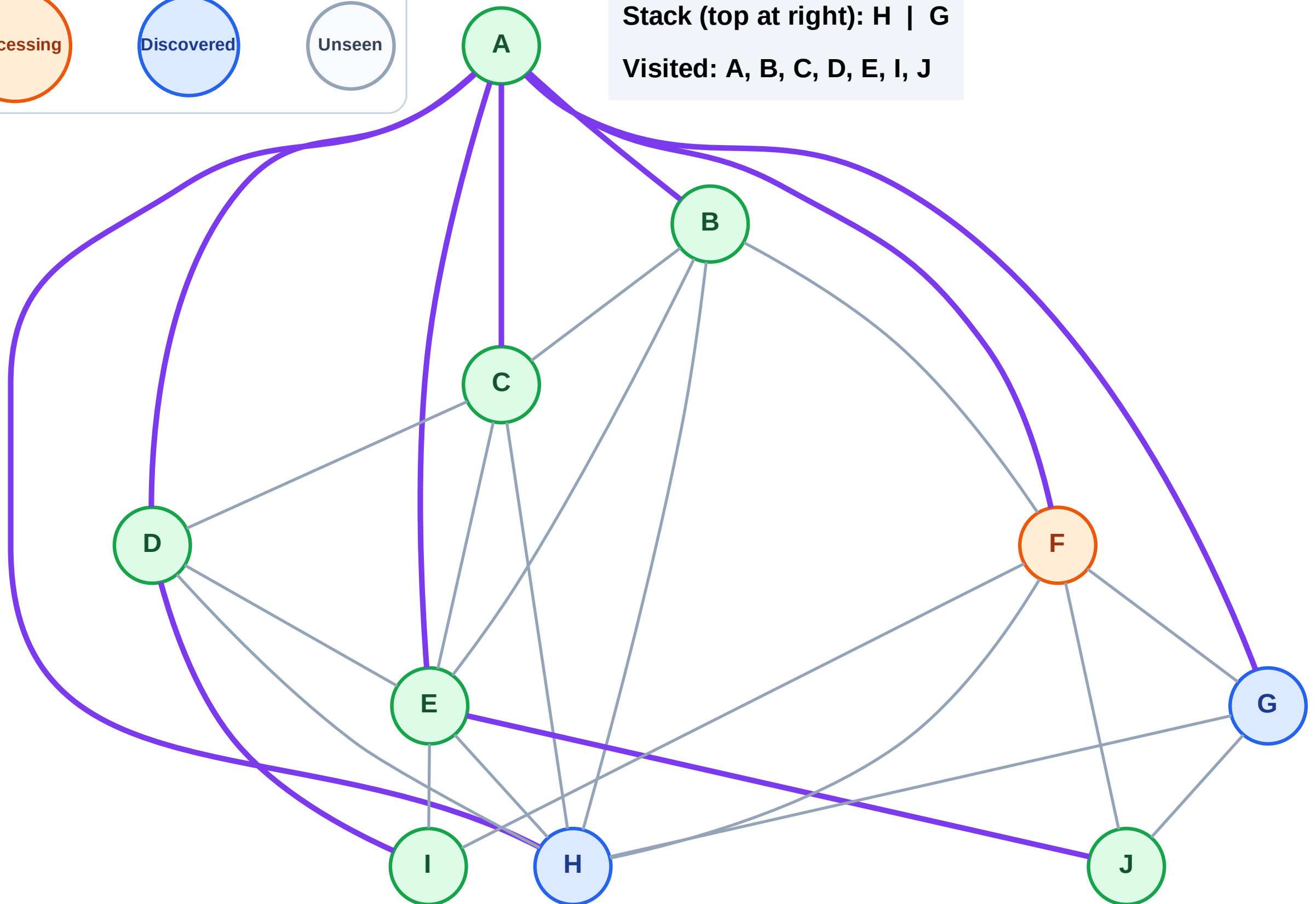
Step 16: Process node F

Legend



Stack (top at right): H | G

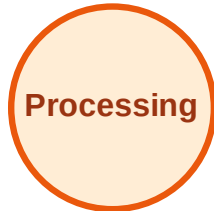
Visited: A, B, C, D, E, I, J



DFS Traversal

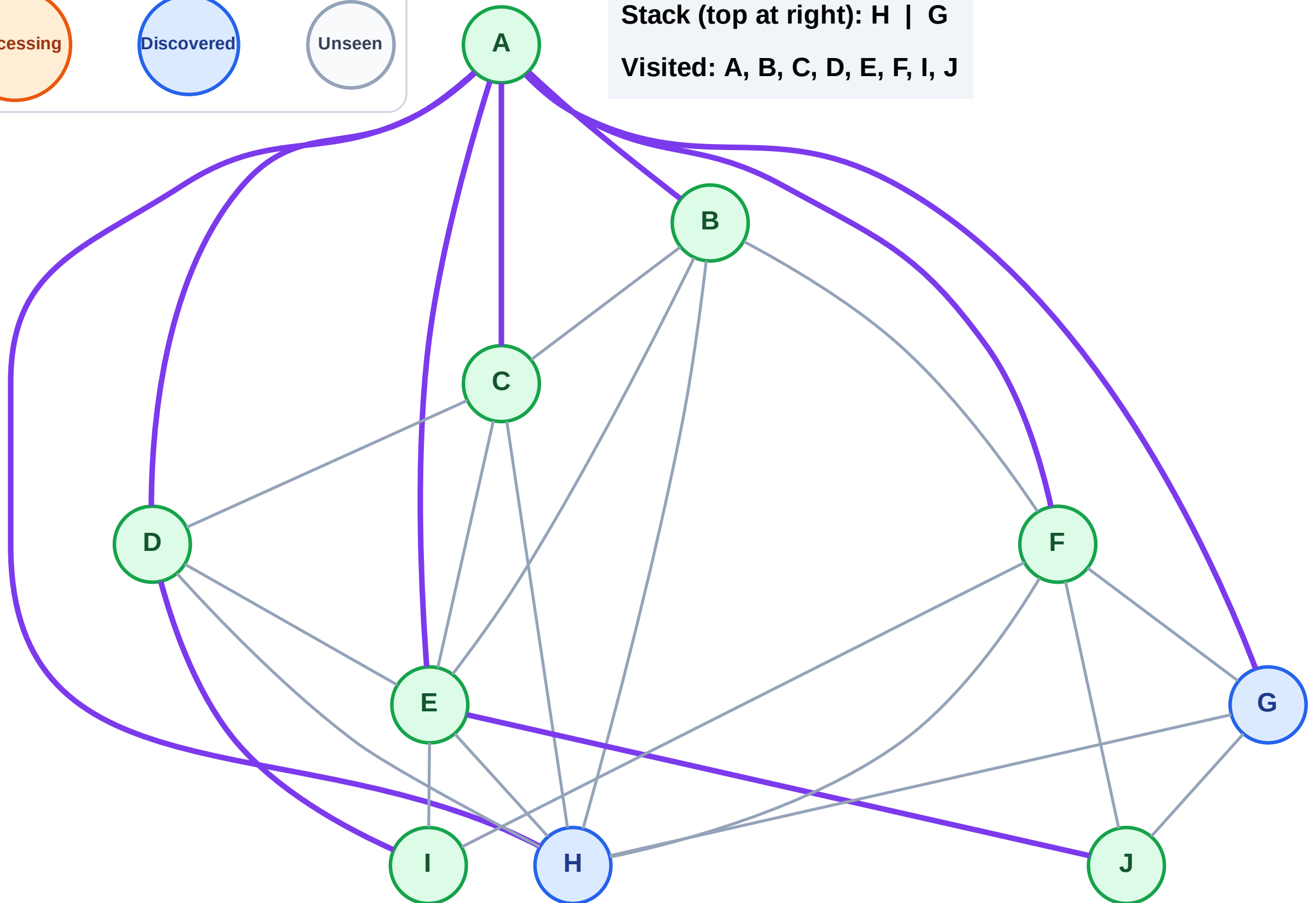
Step 17: No new neighbors from F

Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

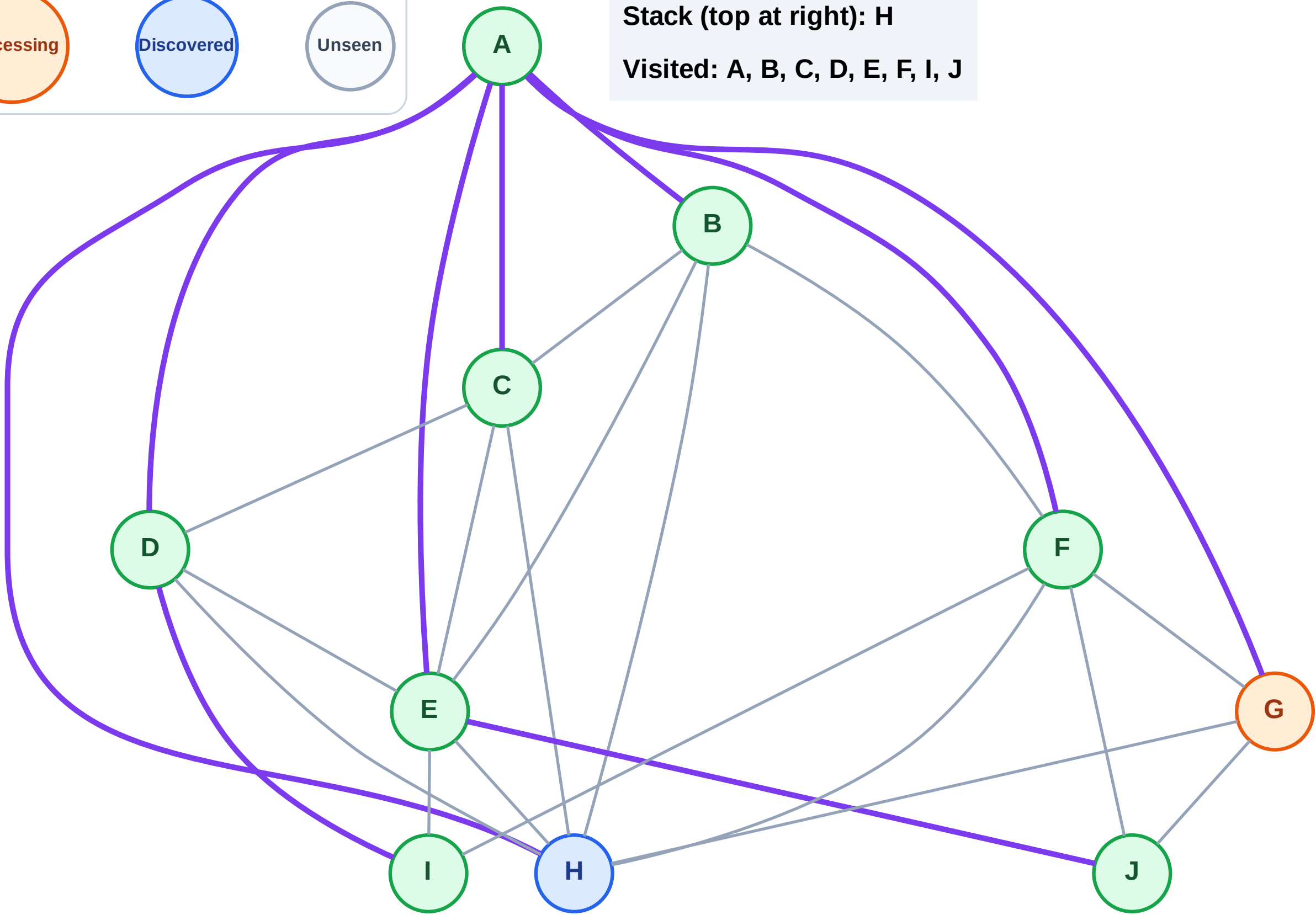
Complete

Processing

Discovered

Unseen

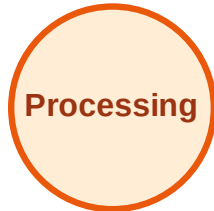
Stack (top at right): H
Visited: A, B, C, D, E, F, I, J



DFS Traversal

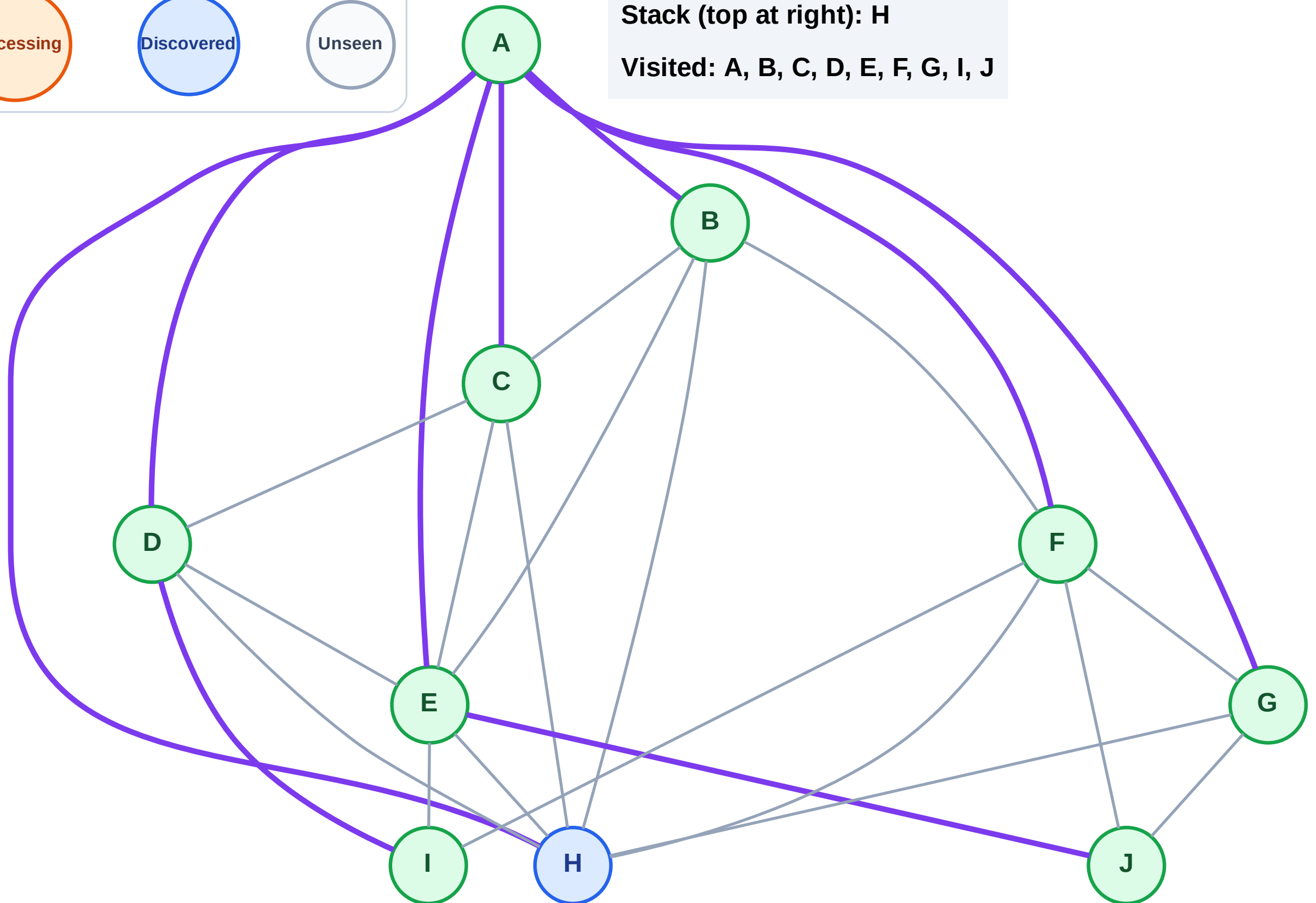
Step 19: No new neighbors from G

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 20: Process node H

Legend

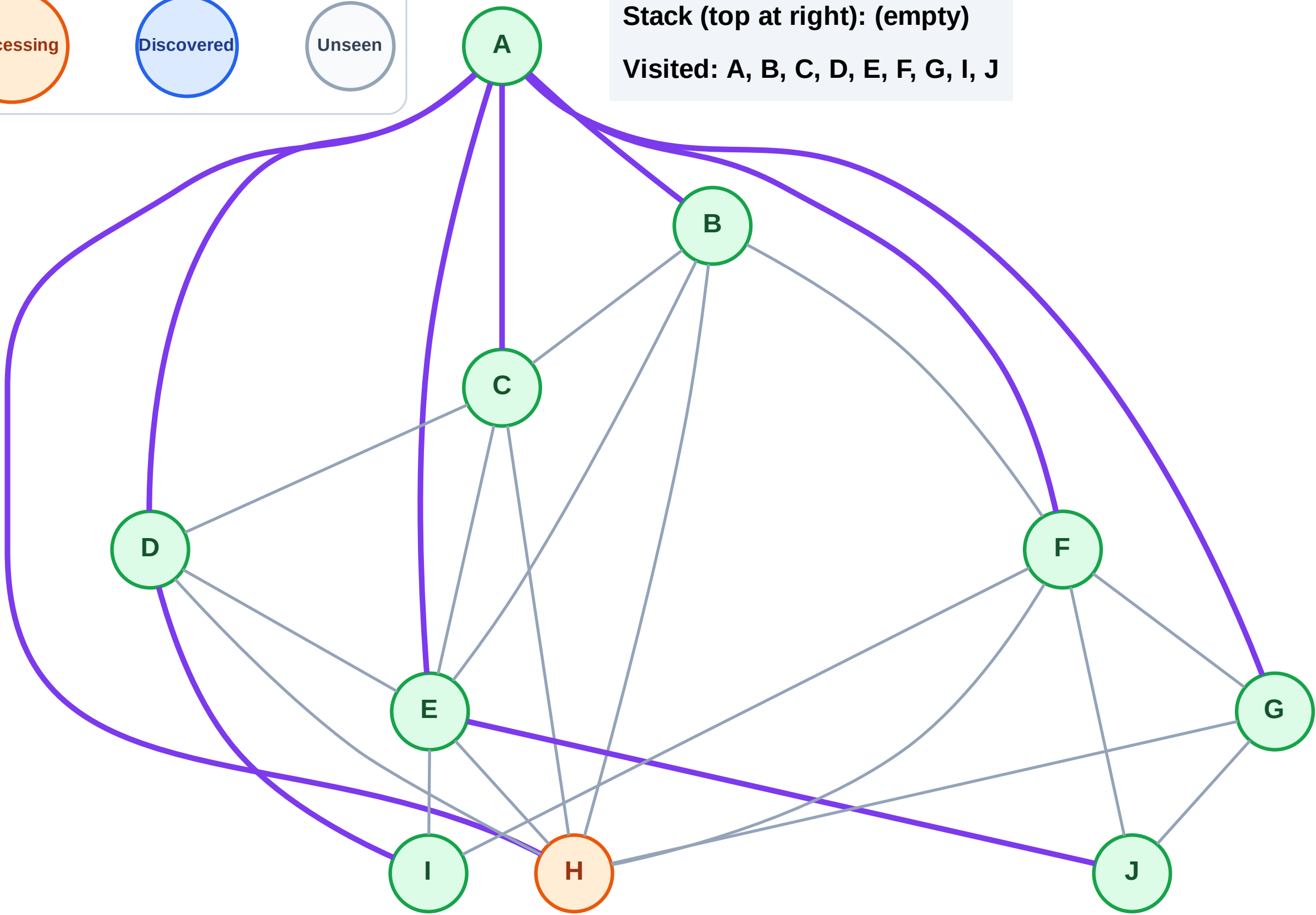
Complete

Processing

Discovered

Unseen

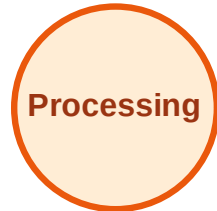
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

